

COVER LETTER OF PAPER 938 $\Rightarrow$ 1358 (SHEPHERDING)

Dear Meta Reviewer and Referees:

We sincerely thank the Meta Reviewer and all three Referees for their positive recommendations and insightful comments throughout the two rounds of review. Following the shepherding decision, we have revised the paper to address the revision items raised in the meta-review, the additional comments from Referee #1, and the minor points from Referee #3. For the convenience of the Referees, changes addressing the concerns of [Meta Reviewer/Referee #1](#) and [Referee #3](#) are highlighted in [cyan](#) and [magenta](#), respectively.

Response to the comments of the Meta Reviewer.

[P1] *The definitions, examples, and descriptions in Section 3 (Preliminaries) are now much more consistent and easier to follow. However, I still find Definition 3.3 somewhat difficult to understand. In particular, what exactly does “or a constant given by the matching tuples of $t_p[X]$ ” mean? (see W1 from previous review)*

[A] Thanks! In Definition 3.3, a CFD $\varphi : (X \rightarrow Y, t_p)$ associates the embedded FD $X \rightarrow Y$ with a pattern tuple t_p over $X \cup \{Y\}$, where the consequent $t_p[Y]$ can be:

(1) a wildcard “_” (variable CFD), in which case any value of Y is allowed and the constraint is purely the embedded FD $X \rightarrow Y$ restricted to tuples matching $t_p[X]$; or

(2) a constant $a \in \text{dom}(Y)$ (constant CFD). In this case, the constraint requires that every tuple $t \in r$ matching the antecedent pattern (i.e., $t[X] \leq t_p[X]$) also takes this consequent value, i.e., $t[Y] = a = t_p[Y]$. The phrase “a constant given by the matching tuples of $t_p[X]$ ” was meant to indicate that this constant is the value consistently shared by the tuples matching the antecedent pattern. For instance, in Example 1, $\varphi_1 : ([\text{Workclass}] \rightarrow \text{Education}, (\text{State-gov} \parallel \text{Bachelors}))$ states that every tuple with $\text{Workclass} = \text{State-gov}$ must have $\text{Education} = \text{Bachelors}$. To remove the ambiguity, we have rephrased Definition 3.3 in the revision (Section 3, p. 3) as: “The consequent $t_p[Y]$ is either a wildcard “_” or a constant $a \in \text{dom}(Y)$. For a constant consequent, every tuple t satisfying $t[X] \leq t_p[X]$ must have $t[Y] = a$. A CFD is called a variable CFD if t_p contains “_”, and a constant CFD otherwise.”

[P2] *The two synthetic datasets are still not sufficiently motivated. What makes them particularly suitable? I understand that they allow characteristics such as noise to be generated in a controlled manner. However, why were these specific characteristics chosen? This does not require a lengthy explanation, but I would expect at least one or two sentences justifying their design. (see D12 from previous review)*

In your response letter, you motivate the synthetic datasets as follows: “While the real-world datasets demonstrate the practical applicability of SCFDM, they do not provide verifiable ground truth CFDs for quantitative evaluation, making it costly to establish ground-truth CFDs due to extensive expert verification and preventing controlled noise injection for robustness evaluation.”

That sounds a little confusing, since you use five real-world datasets in your quantitative evaluation. Does this mean that their ground truth is unreliable? If so, the corresponding experiments would appear to have rather limited value. Or were you only able to establish ground truth for these five datasets and would have liked to include additional

datasets?

[A] Thanks! We clarify that the ground truth of the five real-world datasets is not unreliable; rather, the two dataset groups serve complementary purposes. For the real-world datasets, we establish the ground truth through exhaustive mining with CTANE and CFDMiner followed by expert verification, a standard workflow for constructing ground truth in rule-mining evaluation [30, 31, 92], and use it for all accuracy evaluation. Their limitation is that the resulting dependencies are data-dependent, making it difficult to control the underlying dependency structure, inject noise, and adjust the dataset size (i.e., the number of rows and columns).

The synthetic datasets are therefore designed to provide controlled tests of the two aspects relevant to SCFDM. We use Bayesian-network generators because they provide a known dependency structure by construction while preserving realistic attribute correlations [92], which are the signals that AttrFinder is designed to capture. Following the noise injection strategy in [14], we inject 5% dependency noise and 5% cluster noise into Syn-1 and Syn-2, respectively. These two noise types have been shown to pose greater challenges to automated rule mining, making them suitable for evaluating robustness [14]. Our results in Exp-3–Exp-5 further demonstrate that SCFDM is robust to both types of noise, consistently discovering rules that closely match the dependency structures of the Bayesian networks.

More specifically, Syn-1 uses more attributes to evaluate SCFDM on larger schemas, whereas Syn-2 uses fewer attributes to cover a smaller schema size. Thus, the two synthetic datasets provide complementary schema sizes rather than simply adding more datasets. The revised Section 9.1 now explains this design choice.

Moreover, Bayesian-network generators allow flexible control over dataset scale, including both the number of tuples and attributes, making them adaptable to different evaluation needs. Small synthetic datasets enable rapid algorithm validation, while large synthetic datasets support stress testing. Different noise types [14] can also be injected in a controlled manner to evaluate robustness.

[P3] *The experimental evaluation is now much more comprehensive. Nevertheless, some choices are still made without being sufficiently justified. I understand that space constraints make it impossible to present all results and that some omitted results are included in the appendix. However, in such cases I would either expect a sentence explaining why the presented results were selected or a statement that the remaining datasets exhibit similar patterns or trends (provided that this is indeed the case). Examples include:*

* *The analysis of varying k is shown only on the Census dataset. Since this variable exhibits particularly pronounced increases, these results are especially interesting. By contrast, the results for varying δ show only minor changes, so in that case a single plot would probably have been sufficient to illustrate the trend.*

* *Why are the datasets used for varying $|r|$ different from those used for varying $|R|$?*

* *Why were these particular four datasets selected for Table 3?*

* *Why were these particular two datasets selected for Table 5?*

Again, this does not require a lengthy discussion, but I would expect one or two sentences justifying these choices, possibly with a reference to additional results in the appendix if not provided already. (see W3 and D15 from previous review)

[A] Thanks! We have added one or two sentences at each of the four places to explain the selection rationale and note consistent trends where applicable. We have also included the previously omitted results in Appendix H for completeness.

(1) *Varying k (Census) and varying δ .* Census was selected for varying $|k|$ because it has the widest attribute range among the datasets on which all baselines complete, making the growth of the search space with $|k|$ pronounced. For $|\delta|$, all algorithms show only minor sensitivity because the confidence threshold has little effect on the candidate space. We therefore present two representative plots (Syn-2 and Adult) to illustrate the flat trend, and clarify in Section 9.2 (Exp-2) that the remaining datasets exhibit the same trend, with full results provided in Appendix H.6 of [1].

(2) *Datasets for varying $|r|$ vs. $|R|$.* The two parameters impose different requirements. To evaluate different attribute sizes $|R|$, we select datasets covering a broad range of attribute counts. Specifically, we use RT (65–85), Flight (9–25), and Adult (6–14) to represent wide, medium, and narrow schemas, respectively. Varying the sampling ratio $|r|$ is better tested on datasets large enough that subsampling materially changes the workload, hence Crop, Flight, and RT. We have added this rationale in Section 9.2 (Exp-2) and noted that the other datasets exhibit consistent trends, with their full plots provided in Appendix H.6 of [1].

(3) *Table 3 (varying $|q|$).* The four datasets (RT, Census, Adult, Syn-1) were selected to cover the two pipelines (SCFDM_{part} on constant CFDs, SCFDM_{all} on both constant and variable CFDs) across one real dataset with a wide schema (RT), one with complex dependencies (Census), one small dataset (Adult), and one synthetic dataset with 5% dependency noise (Syn-1), where all configurations complete within the time limit. A sentence explaining this selection rationale has been added to the paragraph on varying $|q|$ in Section 9.2, with full results provided in Appendix H.6 of [1].

(4) *Table 5 (Exp-4).* RT and Syn-1 were chosen as one real-world and one synthetic representative with complementary characteristics. RT has a wide schema (85 attributes) among real datasets, while Syn-1 has designed ground truth. Together, they evaluate whether SCFDM can maintain both precision and recall when the support threshold is low. The complete sensitivity results on the other five datasets are already provided in Appendix H.2 of [1] (as referenced below Table 5), which show consistent trends. We have made this justification explicit in Section 9.2 (Exp-4).

Many thanks for your support and insightful suggestions!

Response to the comments of Referee #1.

Thanks! We note that the three revision items raised by Referee #1 in this round, namely Definition 3.3 ([P1], corresponding to [W1] in the previous round), the motivation for the synthetic datasets ([P2], corresponding to [D12]), and the justification of the experimental choices ([P3], corresponding to [W3] and [D15]), were also highlighted by the Meta Reviewer. To avoid duplication, we provide our detailed responses to these three points in the Meta Reviewer section above. Below, we address the additional minor comments ([R1]–[R14]) from Referee #1.

[R1] *I would not include the discussion of pattern tableau as part of Definition 3. Instead, I would present it as regular text following the*

definition.

[A] Thanks! We have moved the discussion of the pattern tableau $\mathcal{T}_p$ out of Definition 3.3. It now appears as regular text immediately after the definition (Section 3, p. 3), where we present the tableau-based CFD $\varphi_T : (\mathbf{X} \rightarrow Y, \mathcal{T}_p)$ and refer to Appendix B for the extension of SCFDM.

[R2] *The variable η is already used in Section 4 (Parallel Discovery of CFDs) before it is introduced in Section 5.1.*

[A] Thanks! The symbol η (the speedup factor in Section 8.2) and the encoding width $\eta = \sum_{i=1}^m c_i$ (first used in the overview of the vectorization, Section 4/5) were used before their formal introduction. We have fixed the forward references: the one-hot encoding width η is now defined at its first occurrence in the overview (Section 4, where the zero-one matrix $M \in \{0, 1\}^{n \times \eta}$ is first mentioned), and the speedup factor ϑ is similarly defined at its first use in Section 8.2. We have also double-checked that all other symbols are introduced before use.

[R3] *In Algorithm 1, k could also be included as an input parameter.*

[A] Thanks! The attention threshold θ_{att} is computed as z/m in Line 14 using the scaling factor z , which is a user-adjustable parameter (see Appendix H.3 and H.4 of [1]). We have added the scaling factor z to the input of Algorithm 1, and unified the threshold to $\theta_{\text{att}} = z/m$ throughout (please also see our response to the comment of [C2] of Referee #3 on unifying z/m vs. k/m).

[R4] *You repeatedly write: “the conditional probability $\forall t_i, t_j \in r_\varphi : \Pr(t_i[Y] = t_j[Y] \mid \bigwedge_{A \in X} t_i[A] = t_j[A]) \geq \delta$.” However, this is not a conditional probability itself, but rather a condition involving a conditional probability. It might be worth rephrasing this to avoid the ambiguity.*

[A] Thanks! We have rephrased all occurrences of this statement in Sections 6.1 and 7 (e.g., Definition 6.1) as follows: “the CFD holds with confidence δ on r_φ if the conditional probability $\Pr(t_i[Y] = t_j[Y] \mid \bigwedge_{A \in X} t_i[A] = t_j[A])$ is at least δ for tuples $t_i, t_j \in r_\varphi$,” thereby separating the probability from the condition imposed on it.

[R5] *When computing precision and recall, you compare the two sets of minimal CFDs, Σ_s and Σ . Since the objective is to find a minimal cover of minimal CFDs, both sets represent minimal covers, right? Is it possible that a set of CFDs has multiple minimal covers? If so, couldn’t the algorithm produce a different minimal cover than the ground truth? In that case, the intersection between Σ_s and Σ could be small even though the algorithm’s output would still be correct.*

[A] Thanks! A set of CFDs may have multiple minimal covers that are syntactically different but semantically equivalent. Thus, in principle, exact set matching could penalize a correct alternative cover. We address this concern by examining how our covers are constructed and where the unmatched CFDs come from.

(1) *Deterministic mining pipeline.* Both our ground truth Σ (obtained through exhaustive mining with CTANE and CFDMiner in Metanome, followed by expert verification) and SCFDM’s local execution (Algorithm 3, Line 6) use the same level-wise enumeration and the same non-redundancy and minimality criteria defined in Section 3. With a fixed search order, redundant-rule removal is deterministic. Thus, the two pipelines do not arbitrarily choose

among different minimal covers, they follow the same construction procedure, and any differences arise from the actual differences in the mined data and execution process.

(2) *Source of unmatched CFDs.* We further examined the CFDs that are not shared by Σ_s and Σ . The missing ground-truth CFDs are attributable to the sampling and partitioning process, which may prevent some valid CFDs from being discovered, while the additional CFDs produced by SCFDM are attributable to noise or local candidates that do not belong to the ground truth. Therefore, the observed differences in Σ_s and Σ , and hence the precision/recall loss, reflect the actual discovery errors targeted by our evaluation rather than arbitrary choices among equivalent minimal covers.

We have added a clarification in Section 9.1 (accuracy metrics). Although minimal covers may not be unique in general, our ground-truth and local mining pipelines use the same deterministic construction procedure. Thus, the unmatched CFDs observed in our experiments arise from the discovery process itself.

[R6] *What happens if LSH sampling produces more than N buckets?*

[A] Thanks! This case is handled naturally by the round-robin structure of RepSampler (Algorithm 2). Before sampling, RepSampler removes any bucket B_i with $|B_i| < \sigma$ (the support threshold), which guarantees that every remaining bucket has size at least σ and thus sufficient scale to support CFDs satisfying the support requirement. If the LSH partitioning produces $k > N$ non-empty buckets, the first while-iteration samples one tuple from each of the first N buckets in ascending order of bucket size and reaches the target $|S| = N$ (Line 11), so the loop terminates. Small buckets are processed first, as they are sorted by ascending size in Line 5. Thus, even when $k > N$, the N smallest buckets, which are most likely to support low-support CFDs, are represented in S . Hence more buckets than N pose no correctness or termination issue. In practice, the case $k > N$ is unlikely, because the LSH buckets are formed from row similarities in M , which tend to group similar tuples together rather than scatter them randomly across many buckets. We have added clarifying sentences about this case in the description of Algorithm 2 (Section 7.2).

[R7] *What maximum CFD size is used in the experiments where this parameter is not varied?*

[A] Thanks! In all experiments where k is not varied, the maximum CFD size is set per consequent attribute via the entropy-driven strategy of Appendix H.3 of [1] (rather than a fixed constant). The attention-mass retention factor γ yields per-attribute size limits k_γ bounded between 2 and 7 attributes in practice (e.g., on Adult and Crop). We have added this statement to the default settings in Section 9.1.

[R8] *In Exp-2, you use $|\sigma|$, $|\delta|$, $|q|$, and $|k|$. Why are absolute value symbols used here? These are not sets. Shouldn't the notation simply be σ , δ , q , and k ?*

[A] Thanks! The absolute-value bars were inherited from the notation $|r|$ (tuple count) and $|R|$ (attribute count), where the bars denote cardinality, but for scalar thresholds such as σ , δ , k , and q they are indeed inappropriate. We have removed the absolute-value symbols and now write $\hat{\sigma}$, δ , k , and q consistently in Section 9.2 (Exp-2) and Figure 4, keeping $|r|$ and $|R|$ only where cardinality is meant.

[R9] *In Exp-5 (Ablation Study of SCFDM), you write: "however, it*

severely degrades precision by introducing semantic noise and spurious CFDs (to 96.5% and 97.8%)". How is this possible? Aren't all discovered CFDs validated against the entire dataset (Algorithm 3, Line 8)? If so, how could the alternative sampling strategies introduce noisy or spurious CFDs?

[A] We thank the Referee for this insightful question, which helped us clarify the distinction between checking a CFD on the original relation and determining whether it belongs to the ground-truth cover. Line 8 of Algorithm 3 checks whether a CFD satisfies the required support and confidence thresholds on the original relation. However, satisfying these thresholds alone does not determine whether the CFD belongs to the ground-truth cover.

(1) Each candidate CFD φ is validated against the complete relation r and discarded if its support is below σ or its confidence is below δ . Since $\delta < 1$ (default 0.95) and real-world data often contains noise, a CFD may satisfy the confidence threshold while reflecting accidental patterns in the data rather than a true dependency.

(2) Precision is computed against the expert-verified gold standard Σ (Section 9.1), which excludes noise-induced and redundant rules. A discovered CFD is therefore a false positive if it is either (i) *noise-induced*, passing the δ -confidence check because $\delta < 1$ tolerates violations from erroneous tuples, or (ii) *globally redundant*, being valid on r but subsumed by a more general rule in Σ . This can occur because Algorithm 3 unions the per-sub-table local covers (Line 9) without global re-minimization, allowing a locally minimal rule to be entailed by rules from other sub-tables or by the cover.

(3) *w/o AttrFinder* (96.5%) removes correlation-guided partitioning, so candidate antecedents are no longer restricted to attributes correlated with the consequent. Mining arbitrary attribute combinations can therefore produce rules that hold on r due to chance co-occurrences in noisy regions, as well as redundant variants subsumed by other rules, consistent with the phenomenon observed for the baselines in Exp-3. *w/o RepSampler* (97.8%) mines the complete noisy relation, allowing isolated noisy tuple groups to satisfy the σ and δ thresholds and produce noise-induced CFDs that pass Line 8 but are excluded from Σ . In contrast, the diversity-aware stratified sampling of RepSampler selects representative tuples and reduces the impact of isolated noise, with recall guaranteed by Theorem 2.

We have clarified the distinction between verification and precision in Section 8.1 and Exp-5. Algorithm 3 (Line 8) verifies candidates against the complete relation r , whereas precision is measured against the ground-truth CFDs Σ established in Section 9.1. Thus, satisfying the support and confidence thresholds on r is required, but a CFD must also pass the ground-truth mining and expert verification process to be included in the ground-truth cover.

[R10] *In Exp-5 (Ablation Study of SCFDM), you write: "(computing pairwise attribute correlation)". It seems that there is either an extra space before the closing parenthesis or some missing words.*

[A] Thanks! We have revised it to "(computing pairwise attribute correlation between all attribute pairs)".

[R11] *In Exp-5 (Ablation Study of SCFDM), you write: "indicates that men around 50 working over 50 hours per week ...". Where does the age restriction "around 50" come from?*

[A] Thanks! The quoted sentence in Exp-6 misread our rule. In the

Adult preprocessing, both Age and Hours are top-coded, with values ≥ 50 recorded as 50. Thus, the antecedent pattern (male, 50, sales || married-civ-spouse) of ϕ_6 means Age ≥ 50 and Hours ≥ 50 , rather than around 50” or over 50 hours.” The rule itself is reasonable: it satisfies the support and confidence thresholds on r , and its semantics align with common intuition, as senior individuals who still work long hours are more likely to be married and bear family-supporting responsibilities. This is exactly the kind of statistically grounded and interpretable CFD that the case study is intended to highlight. We have therefore revised the sentence to: “ ϕ_6 indicates that men aged 50 or above who work 50 or more hours per week in sales are likely to have Marital = married-civ-spouse.”

[R12] *What exactly does SCFDM without PCFDFinder do? Are all sub-tables simply processed sequentially in this variant?*

[A] Thanks! In the w/o PCFDFinder variant, the pipeline (encoding, AttrFinder, and RepSampler) remains unchanged, but the generated sub-tables are processed sequentially by the mining engine on a single thread. We disable the multi-threaded execution of Algorithm 3 (Lines 4–5) while retaining its partitioning, verification, and union steps. As a result, this variant achieves the same accuracy as the full model (98.6% precision, 97.9% recall, and 98.2% F1-score in Table 6), while runtime increases from 655.4s to 18,621.3s, showing the contribution of parallelism. We have clarified this setup in Exp-5.

[R13] *It would also be helpful to see quantitative results for Exp-6 (even if only in the appendix). How many erroneous values were corrected? How many correct values were incorrectly modified? In particular, CFD ϕ_6 seems as it could alter some correct values into incorrect ones, since not all men working more than 50 hours per week in Sales are married. For those who are not, the repair process would actually introduce new errors. Without supporting quantitative results, I find the statement “The case study shows that SCFDM discovers high-quality CFDs for repairing noisy data” somewhat difficult to justify.*

[A] Thanks! We agree that quantitative repair results are needed to substantiate this claim. We have therefore added a quantitative evaluation in the new Appendix H.5 of [1] using the noisy Adult dataset, where ground-truth errors are known by construction. We injected errors into 1,000 cells, including mislabeled gender values, missing values, and corrupted marital-status entries. Applying the discovered CFDs, including ϕ_5 and ϕ_6 , as repair rules corrected 912 of the 1,000 errors, achieving 94.6% repair precision and 91.2% recall. The overall repair performance achieves an F_1 score of 0.929. For ϕ_6 , which matches Hours = 50 exactly, only 31 repairs were triggered, of which 28 were correct and 3 introduced new errors, consistent with its 0.97 confidence on the clean reference. We have also revised the claim in Exp-6 to “the case study and the quantitative repair evaluation (Appendix H.5) show that SCFDM discovers high-quality CFDs that support effective data repair.”

[R14] *In Table 5, it would also be helpful to mark the runs that exceeded the time limit with an asterisk, as is done in Table 4.*

[A] Thanks! We have marked the timed-out entries in Table 5 with an asterisk (*), consistent with the convention of Table 4. The explanatory footnote of Table 4 has been extended to Table 5 as well.

Many thanks for your support and insightful suggestions!

Response to the comments of Referee #2.

Many thanks for your support and insightful suggestions! We appreciate that there are no further concerns in this round, and we are grateful for your constructive feedback in the previous round, which helped us further improve the clarity and quality of the paper.

Response to the comments of Referee #3.

[C1] *Definition 3.5 seems not connected to what was written before or after. Maybe consider moving it to section 4, where it is needed. Also Definition 3.5 is referring to the correlated attribute set as C_Y , later it is referred to as $Corr_Y$. That should be unified.*

[A] Thanks! We agree that Definition 3.5 was isolated in Section 3, and have moved it to Section 4 (solution overview), where the correlated attribute set is first used to describe the partitioning module. Following the unification request, we now consistently denote the correlated attribute set of a consequent Y as $Corr_Y$ in definitions and in the overview, and denote their collection as $Corr = \{Corr_Y \mid Y \in attr(\mathcal{R})\}$ in Sections 6.3 and 8. We have double-checked all occurrences for consistency.

[C2] *In Section 6.3 Step 2 θ_{att} is set as z/m , later in example 4 as k/m and in Algorithm 1 also as k/m . This should also be unified.*

[A] Thanks! The letter k in Example 4 and Algorithm 1 was a leftover that conflicts with the maximum CFD size k used in the experimental section, while denoting the same scaling factor as z in Section 6.3. We have unified the notation as $\theta_{att} = z/m$ throughout Section 6.3, Algorithm 1, and Example 4, where the threshold is now $3/14$ with $z = 3$ and $m = 14$. We also declare z as an input to Algorithm 1 (please also see our response to **[R3]** of Referee #1).

[C3] *Also while I’m glad that the number of attention heads was added in section 6.2, the sentence seems somehow unconnected. I would be happy if that could be rephrased slightly.*

[A] Thanks! We have rephrased the sentence in Section 6.2 and integrated it into the surrounding discussion: “Stacking L layers ($L = 6$ as commonly used for tabular data [42, 45]) enables iterative refinement of attribute dependencies, e.g., capturing cases where X_1 and X_2 jointly determine Y . To capture dependencies among multiple attributes, the multi-head self-attention uses four attention heads, whose attention maps are averaged in Section 6.3.”

Many thanks for your support and insightful suggestions!

Once again, we sincerely thank the Meta Reviewer and all Referees for the constructive comments that have substantially improved the quality of this paper.

Fast Discovery of Conditional Functional Dependencies via Transformer-Guided Relation Partitioning

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ABSTRACT

Discovering conditional functional dependencies (CFDs) in large relations has been a long-standing challenge due to prohibitive computational costs. We propose SCFDM, a Transformer-guided framework that accelerates the discovery by modeling CFDs as latent conditional distributions. By projecting relational data into a high-dimensional space via a Transformer-based model, SCFDM leverages self-attention saliency to identify correlated attributes by capturing their interaction strengths. This neural-driven insight allows SCFDM to divide the exponentially large search space into independent smaller ones. To ensure discovery quality, we propose a diversity-aware sampling strategy with a theoretical lower bound for discovery accuracy, which reduces input size with performance guarantee of CFD discovery. This Transformer-guided division of the input relation enables data parallelism for any existing CFD discovery algorithm. Extensive experiments on seven real-world and synthetic datasets demonstrate that SCFDM achieves an average speedup of 105.18 $\times$ (up to 136.49 $\times$) compared with baselines while maintaining high discovery accuracy of over 90%.

1 INTRODUCTION

Data dependencies, such as conditional functional dependencies (CFDs) [29], matching dependencies (MDs) [27] and functional dependencies (FDs) [19], are crucial in data management tasks, including query optimization [52], data cleaning [28], and data repair [28]. However, dependency discovery remains challenging, especially for those that contain patterns such as CFDs [31]. It is costly to discover CFDs, as it involves both identifying FDs embedded in CFDs and mining associated patterns [41], which takes $O(m \times d^m \times n)$ [73], where m , n and d are the number of attributes, tuples, and average domain size of attributes, respectively. To accelerate CFD discovery, researchers have proposed various algorithms [16, 18, 24, 25, 30, 40, 41, 47, 50, 55, 56, 73, 90, 94] with optimization strategies as follows. (1) Pruning strategies eliminate invalid candidates early to reduce the search space [30, 55, 73, 94]. (2) Tuple sampling selects subsets from the original relation to handle large datasets [8, 10, 31, 53, 58]. (3) Parallel mining employs multi-processor processing to accelerate the discovery [31, 32, 70].

However, previous methods still face the following challenges in practice. First, most existing methods discover directly on the input relation by level-wise search, where the exploration space grows exponentially with the number of attributes [30, 67]. This incurs unbearable enumeration and high memory cost when the input scales in columns. Although the support-based pruning [30–32] helps to alleviate this problem, the discovery can hardly terminate

in practice when a low support threshold is set. Second, existing parallel paradigms suffer from excessive synchronization overhead as they manage interdependent search spaces [31]. As the pruning and validation of rules often depend on the state of other processors, the parallel threads frequently become idle while waiting for synchronization, resulting in high communication cost to maintain consistency across multiple cores [54]. Such dependencies among the cores make it difficult to achieve linear speedup and optimal load balancing [31]. Third, although most existing sampling methods (e.g., single-round or reservoir sampling) operate within accuracy bounds [10, 13, 31, 93], they treat each tuple equally when sampling and ignore the semantic correlations among them. Thus, these sampling methods cannot skip unrelated rows in which CFDs are unlikely to be discovered and may miss relatively rare tuples where low-support CFDs are embedded, which cause redundant search and fail to discover valid CFDs of low support.

These challenges lead to the following questions: (1) Can we capture correlations among attributes before discovery to prune the search space? (2) How can we sample tuples such that the majority of CFDs, including low-support ones, can be discovered from the sampled data? (3) Can we partition the input relation into independent sub-tables based on attribute correlations to accelerate CFD mining via data parallelism? (4) Given noisy real-world datasets, can we discover CFDs that are valid for data management tasks?

Contributions & organization. In view of these challenges, we propose SCFDM, a Sub-table-based Conditional Functional Dependency data-parallel Mining framework that accelerates CFD discovery via data parallelism where the input relation is divided into independent sub-tables. Specifically, each sub-table is constructed by grouping correlated attribute sets identified by a Transformer-based model and sampling similar tuples via a representative sampling strategy. SCFDM allows any existing CFD discovery algorithms to be executed on these sub-tables in parallel, which significantly accelerates the mining process while maintaining accuracy. SCFDM has the following features.

(1) *Vector representation for CFD discovery (Section 5).* We develop a two-stage embedding pipeline that transforms raw relations into vector representations for correlation learning. We first employ one-hot encoding to convert discrete categorical values into a sparse zero-one matrix M , which is subsequently utilized to identify similar tuples that support CFDs. Vectors in M are then projected into a dense latent space to construct the input tensor $\mathcal{T}$, enabling the Transformer-based model to capture attribute-level correlations and identify the correlated attribute sets required for mining CFDs. (2) *Probabilistic modeling of CFDs (Section 6).* We define CFDs from a statistical perspective, which formulates attribute depen-

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dencies as conditional probability distributions. By approximating these distributions, the Transformer-based model can effectively capture and learn the correlations of attributes in CFDs.

(3) Attention-driven correlated attribute set extraction (Section 6).

We propose AttrFinder, a Transformer-guided module that extracts correlated attribute sets by analyzing the Transformer’s global attention distributions. Instead of enumerating in the exponentially large search space, SCFDM identifies the correlations among attributes by analyzing the attention maps of the Transformer. This approach extracts highly correlated attribute sets from the learned attention weights, which are then used to partition the input relation into independent sub-tables. These sub-tables enable efficient data parallelism for existing CFD discovery algorithms.

(4) Diversity-aware representative tuple sampling (Section 7). To reduce computational overhead while providing a discovery recall guarantee, SCFDM establishes a theoretical sampling lower bound. By incorporating binomial variance into the probabilistic analysis of CFD support, SCFDM derives a tighter sample budget compared to standard Chernoff bounds. To populate this budget, we employ diversity-first selection guided by locality sensitive hashing (LSH), which partitions tuples into similarity-aware buckets. This ensures uniform sampling coverage across all data buckets, which prevents low-support CFDs from being obscured by frequent patterns while maintaining the diversity of CFDs within the sampling bound.

(5) Parallel discovery of CFDs (Section 8). SCFDM discovers CFDs through data parallelism guided by correlated attribute sets and representative tuples, where sampled representative tuples are projected onto correlated attribute sets to partition the original relation into smaller independent sub-tables that serve as parallel inputs.

(6) Experimental study (Section 9). Using seven real-world and synthetic datasets, we validate the following results: (a) SCFDM speeds up classical CFD mining algorithms by 105.18 times on average, while maintaining over 90% accuracy. (b) SCFDM exhibits favorable scalability as the dataset dimension grows. For example, as the number of attributes increases from 9 to 25 on the Flight dataset, its runtime rises by 31.24× only, compared to a 330.50× increase in the runtime of CTANE [30]. (c) SCFDM scales effectively with more processing cores, achieving a 2.41× speedup on the RT dataset when the number of cores increases from 28 to 112.

Novelty. The novelty of SCFDM includes: (1) a probabilistic formulation of CFDs as latent conditional distributions via a Transformer; (2) an attention-driven module for extracting correlated attribute sets and reducing the search space; (3) a diversity-aware sampling strategy with a theoretical lower bound; and (4) a data-parallel framework that constructs sub-tables via correlated attribute sets and representative sampling tuples for parallel CFD mining.

2 RELATED WORK

CFD discovery. Researchers have proposed various CFD discovery algorithms as follows: (1) Level-wise search with pruning is widely employed for discovering CFDs, including CTANE [30], constant-pattern-generation-based [41, 82], attribute-grid-based [16], FACD [55] and CFUN [24]. (2) Depth-first search. FastCFD [30] uses a depth-first strategy with closed-itemset pruning, while BFCFD [56] employs sampling for single-pass discovery.

(3) Hybrid approaches. As shown in [73], these methods first mine patterns from itemsets and then discover FDs in the corresponding subsets, or first discover FDs and then mine patterns applicable to those FDs. (4) Others. A greedy approximation algorithm computes near-optimal tables of CFDs from embedded FDs [41]. CFDMiner employs generators and closures for efficient constant CFD discovery [30]. GCDF discovers CFDs by detecting data errors guided by user-annotated values [50]. XPlode identifies the top-ranking CFDs for data repair tasks using an on-demand approach [72].

Different from previous methods, SCFDM formulates attribute dependencies in CFDs as conditional distributions learned by a Transformer. By analyzing self-attention saliency maps, SCFDM extracts correlated attribute sets for parallel discovery. These sets partition the original relation into lower-dimensional sub-tables, effectively reducing the exponential search space into smaller ones. This Transformer-guided strategy enables fast CFD discovery via data parallelism and can be integrated with existing discovery methods.

Tuple sampling. Tuple sampling has been applied to dependency discovery [2, 9, 30, 56, 68, 69, 83, 84], including random sampling [63], stratified sampling [34], reservoir sampling [88], and focused sampling [7]. (1) Random sampling selects tuples with equal probability, where Chernoff [93] and Chebyshev [58] bounds are employed to calculate sample size based on error and confidence thresholds. (2) Stratified sampling divides data into multiple sets before sampling, which reduces the sample size based on data distribution and transactions [57, 91], where the main challenge is how to partition the strata [65]. (3) Reservoir-sampling-based CFD discovery iteratively samples data and refines the discovered rules. During this process, each tuple is evaluated to determine whether its attribute values satisfy the rules [56]. (4) Focused sampling targets at tuple pairs specified by consistency sets, evidence sets [10, 53], or errors in user-annotated attributes [50].

SCFDM differs from previous methods in that it not only samples tuples but also divides the input relation by selected columns. Specifically, based on the attention mechanism of Transformers, it reduces the search space by projecting the input onto correlated attributes where candidate CFDs are embedded. Moreover, theoretical analysis provides performance guarantee of support and confidence bounds for the sampling strategy in SCFDM.

Parallel dependency discovery. Various parallel optimizations for dependency discovery have been proposed. ParaCoDe uses a depth-first strategy with a single-node shared-everything architecture [78] to mine CFDs in parallel within a partitioned search tree, which may however suffer from node overloads [37]. Saxena et al. introduce six primitives to parallelize dependency discovery, which focus on workload allocation [48]. However, these primitives do not support CFD patterns [31]. Naumann et al. parallelize approximate dependency mining by traversing multiple search spaces [53]. Han et al. propose parallel rule discovery algorithms that ensure scalability with consideration of computational overhead [31, 32]. However, as data scales, these algorithms suffer from communication and task-allocation bottlenecks due to static coupling between search spaces and resources, limiting dynamic load balancing.

Different from previous parallel solutions [48, 53, 78], SCFDM provides a data-parallel framework that can be integrated with any existing CFD mining algorithm. By discovering CFDs within all sub-

Table 1: Example relation of adult information

ID	Name	Age	Workclass	Education	Edu-num	Relationship	Gender	Occupation	Cc	Ac	City	Proto	Service
t_1	Andy	23	Self-emp	Bachelors	13	Not-in-family	Male	Exec-managerial	05	18	LD	TCP	MQTT
t_2	Edward	34	State-gov	Bachelors	13	Husband	Male	Adm-clerical	01	33	NY	UDP	DNS
t_3	Tom	25	Private	HS-grad	9	Husband	Male	Handlers-cleaners	08	22	MS	TCP	MQTT
t_4	Angelia	33	Private	Masters	14	Wife	Female	Exec-managerial	05	18	LD	UDP	DNS
t_5	Justin	36	Self-emp	Masters	14	Husband	Male	Exec-managerial	08	22	MS	TCP	MQTT
t_6	Alex	42	State-gov	Bachelors	13	Not-in-family	Male	Prof-specialty	01	33	NY	UDP	DNS
t_7	Candice	29	State-gov	Bachelors	13	Wife	Female	Prof-specialty	01	33	NY	TCP	MQTT

tables, the rule discovery problem is deconstructed into independent search tasks. This data parallelism allows SCFDM to inherit existing well-designed search optimizations (e.g., pruning and auxiliary data structures in [30]) and the advanced performance of previous serial methods without altering their mining algorithms.

3 PRELIMINARIES

Consider a relational schema $\mathcal{R}$ defined over a set of attributes $\text{attr}(\mathcal{R})$, where each attribute $A \in \text{attr}(\mathcal{R})$ has a domain $\text{dom}(A)$. For each tuple t in relation r of $\mathcal{R}$, $t[A] \in \text{dom}(A)$, where $t[A]$ is the value of t for attribute A . Next, we review the basics of CFDs.

Definition 3.1: As in [30], the pattern tuple t_p is defined over the attributes $\mathbf{X} \cup \{Y\}$, where for each attribute $A \in \mathbf{X} \cup \{Y\}$, $t_p[A]$ is either a constant $a \in \text{dom}(A)$ or an unnamed variable “_” that is a wildcard representing any value from $\text{dom}(A)$.

Definition 3.2: Following [30], we use the concatenation symbol “||” to partition the antecedent and consequent attribute values within the pattern tuple, denoted as $t_p = (t_p[\mathbf{X}] \parallel t_p[Y])$.

Definition 3.3: Given a relation r over $\mathcal{R}$, a set of attributes $\text{attr}(\mathcal{R})$, and a pattern tuple t_p , a conditional functional dependency φ is defined as:

$$\varphi : (\mathbf{X} \rightarrow Y, t_p),$$

where $\mathbf{X}$ is a set of attributes in $\text{attr}(\mathcal{R})$, Y is an attribute in $\text{attr}(\mathcal{R})$, and $\mathbf{X} \rightarrow Y$ denotes the functional dependency embedded in φ . The consequent $t_p[Y]$ is either a wildcard “_” or a constant $a \in \text{dom}(Y)$. For a constant consequent, every tuple t satisfying $t[\mathbf{X}] \leq t_p[\mathbf{X}]$ must have $t[Y] = a$. A CFD is called a variable CFD if t_p contains “_”, and a constant CFD otherwise.

Following [20, 29, 41], a CFD with a pattern tableau is $\varphi_{\mathcal{T}} : (\mathbf{X} \rightarrow Y, \mathcal{T}_p)$, where the tableau $\mathcal{T}_p = \{t_{p_1}, \dots, t_{p_m}\}$ is a finite set of pattern tuples and a tuple $t \in r$ satisfies $\varphi_{\mathcal{T}}$ if t matches at least one $t_{p_i} \in \mathcal{T}_p$. We discuss in Appendix B how to extend SCFDM to such tableau-based CFDs with few modifications.

Semantics of CFDs. Following [30], an order $\leq$ on constants and the unknown variable “_” is defined as follows: $\eta_1 \leq \eta_2$ if $\eta_1 = \eta_2$, or η_1 is a constant and η_2 is “_”. The order $\leq$ naturally extends to tuples, e.g., $(\text{Male}, \text{UDP} \parallel _) \leq (_, \text{UDP} \parallel _)$. $t_{p1}[\mathbf{X}] \ll t_{p2}[\mathbf{X}]$ if $t_{p1}[\mathbf{X}] \leq t_{p2}[\mathbf{X}]$ but $t_{p2}[\mathbf{X}] \not\leq t_{p1}[\mathbf{X}]$, i.e., t_{p2} is more general than t_{p1} . For example, $(\text{Male}, \text{UDP} \parallel _) \ll (_, \text{UDP} \parallel _)$. Given a CFD $\varphi : (\mathbf{X} \rightarrow Y, t_p)$ and an instance r of $\mathcal{R}$, we say that r satisfies φ , denoted by $r \models \varphi$, iff for each pair of tuples t_1, t_2 in r , if $t_1[\mathbf{X}] = t_2[\mathbf{X}] \leq t_p[\mathbf{X}]$, then $t_1[Y] = t_2[Y] \leq t_p[Y]$.

Example 1: We illustrate the syntax and semantics of both constant and variable CFDs based on tuples in Table 1.

Syntax: Consider the following three CFDs defined over Table 1: $\varphi_1 : ([\text{Workclass}] \rightarrow \text{Education}, (\text{State-gov} \parallel \text{Bachelors}))$, $\varphi_2 : ([\text{Education}] \rightarrow \text{Edu-num}, (_ \parallel _))$, and $\varphi_3 : ([\text{Gender}, \text{Proto}] \rightarrow \text{City}, (_, \text{UDP} \parallel _))$. φ_1 is a constant CFD as its pat-

tern tuple contains only specific constants. It dictates that the functional dependency $\text{Workclass} \rightarrow \text{Education}$ is restricted to the pattern $(\text{State-gov} \parallel \text{Bachelors})$. φ_2 and φ_3 are variable CFDs since they contain the wildcard “_” that can take any value in the attribute domains.

Semantics: A relation conforms with a CFD if any pair of tuples matching the antecedent pattern adheres to the consequent pattern. For the constant CFD φ_1 , we filter tuples where $\text{Workclass} = \text{State-gov}$ (i.e., t_2, t_6, t_7). Since all of them have $\text{Education} = \text{Bachelors}$, Table 1 satisfies φ_1 . For the variable CFD φ_3 , it applies to all tuples whose $\text{Proto} = \text{UDP}$ regardless of their gender and city (due to the wildcard “_”). The matching tuples are t_2, t_4, t_6 , all of which satisfy the embedded FD $[\text{Gender}, \text{Proto}] \rightarrow \text{City}$ in φ_3 , meaning Table 1 satisfies φ_3 .

By contrast, Table 1 does not satisfy the variable CFD $\varphi : ([\text{Cc}] \rightarrow \text{Proto}, (_ \parallel _))$, as t_1 and t_4 violate its semantic. Specifically, although both tuples project to the same value on the antecedent attribute ($t_1[\text{Cc}] = t_4[\text{Cc}] = 05$), they map to conflicting values on the consequent attribute ($t_1[\text{Proto}] = \text{TCP} \neq t_4[\text{Proto}] = \text{UDP}$).

Support. Following previous CFD discovery methods [30, 41, 50, 51], the support of a CFD $\varphi = (\mathbf{X} \rightarrow Y, t_p)$ in a relation r , denoted by $\text{supp}(\varphi, r)$, is defined as the number of tuples that match the pattern t_p . Specifically, let $r_{\varphi} = \{t \in r \mid t[\mathbf{X}] \leq t_p[\mathbf{X}] \wedge t[Y] \leq t_p[Y]\}$, the support of φ is given by $\text{supp}(\varphi, r) = |r_{\varphi}|$. φ is σ -frequent if $\text{supp}(\varphi, r) \geq \sigma$, where σ is the support threshold.

Confidence. We adopt the g_3 -based confidence measure [20, 41, 42, 50] to evaluate CFD reliability in the presence of data inconsistencies. Let $r_{\varphi} = \{t \in r \mid t[\mathbf{X}] \leq t_p[\mathbf{X}] \wedge t[Y] \leq t_p[Y]\}$. The confidence of φ on r , denoted by $\text{conf}(\varphi, r)$, is the maximum fraction of tuples in r_{φ} that satisfy the embedded FD $\mathbf{X} \rightarrow Y$ [20]. Formally, $\text{conf}(\varphi, r) = \frac{\max\{|s| \mid s \subseteq r_{\varphi} \wedge s \models (\mathbf{X} \rightarrow Y)\}}{|r_{\varphi}|}$. A CFD φ is δ -confident if $\text{conf}(\varphi, r) \geq \delta$, where δ is the confidence threshold.

Definition 3.4: Following Definition 3.3, given the support and confidence thresholds σ and δ , and a relation r , a CFD φ over r that satisfies σ and δ is defined as:

$$\varphi : (\mathbf{X} \rightarrow Y, t_p), \text{ s.t. } \text{supp}(\varphi, r) \geq \sigma \wedge \text{conf}(\varphi, r) \geq \delta.$$

The confidence threshold helps to mine CFDs in real-life data with noise, where CFDs that violate a few noisy tuples can still be discovered. Meanwhile, the support threshold filters out trivial or complex CFDs that overfit noisy data [92], since support is anti-monotonic and prunes invalid CFDs during rule discovery [30].

Example 2: Continue with Example 1, given a CFD $\varphi_4 : ([\text{Gender}] \rightarrow \text{Relationship}, (_ \parallel _))$, we have $\text{supp}(\varphi_2, r) = \text{supp}(\varphi_4, r) = 7$ (supported by all tuples in r), $\text{supp}(\varphi_1, r) = 3$ (supported by $\{t_2, t_6, t_7\}$), $\text{conf}(\varphi_3, r) = 1$ (supported by $\{t_2, t_4, t_6\}$ without any violations), and $\text{conf}(\varphi_4, r) = \frac{5}{7}$ (violated by $\{t_1, t_6\}$).

Minimality of CFDs. A CFD $\varphi : (\mathbf{X} \rightarrow Y, t_p)$ over r is nontrivial if $Y \notin \mathbf{X}$. Under the support (σ) and confidence (δ) thresholds, (1) a constant [30] CFD $\varphi : (\mathbf{X} \rightarrow Y, t_p)$ is left-reduced over r if for any subset $\mathbf{X}' \subset \mathbf{X}$, $r \not\models (\mathbf{X}' \rightarrow Y, t_p)$; (2) a variable [30] CFD

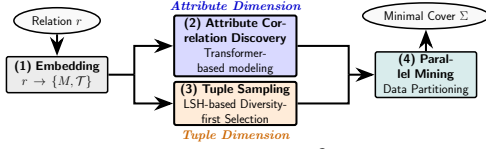


Figure 1: Overview of SCFDM

$\varphi : (X \rightarrow Y, t_p)$ is *left-reduced* over r if $r \not\models (X' \rightarrow Y, t_p)$ for any subset $X' \subset X$, and $r \not\models (X \rightarrow Y, t'_p)$ for any t'_p with $t_p \ll t'_p$. A minimal CFD φ over r is *nontrivial* and *left-reduced*.

Example 3: We illustrate the minimality using Table 1. First, consider a constant CFD $\varphi_c : ([\text{Workclass}, \text{Gender}] \rightarrow \text{Education}, (\text{State-gov}, \text{Male} \parallel \text{Bachelors}))$. While φ_c holds, it is not left-reduced (hence non-minimal) as the attribute Gender is redundant; the subset $X' = \{\text{Workclass}\}$ forms the minimal rule φ_1 in Example 1. Second, consider a variable CFD $\varphi_v : ([\text{Education}] \rightarrow \text{Edu-num}, (\text{Bachelors} \parallel 13))$. Although φ_v is nontrivial, it is not minimal since its pattern can be generalized [30] by replacing the constant with a wildcard, yielding a more general minimal rule $\varphi_2 : ([\text{Education}] \rightarrow \text{Edu-num}, (_ \parallel _))$ that covers all tuples.

Cover of CFDs. Consider a set Σ of minimal CFDs on r [31]. We say that Σ entails a CFD φ , denoted by $\Sigma \models \varphi$, if $r \models \Sigma$ deduces $r \models \varphi$. Σ is *equivalent* to another set Σ' , denoted by $\Sigma \equiv \Sigma'$, if $\Sigma \models \varphi'$ for any $\varphi' \in \Sigma'$ and vice versa. Σ is *minimal* if for each CFD $\varphi \in \Sigma$, Σ is not equivalent to $\Sigma \setminus \varphi$, i.e., each CFD φ in Σ is *non-redundant*. A cover Σ_c of Σ on r is a subset of Σ such that (1) $\Sigma_c \equiv \Sigma$, (2) all rules φ in Σ_c are *minimal*, and (3) Σ_c is minimal, i.e., Σ_c includes no redundant rules.

Problem statement. Given a noisy relation r of schema $\mathcal{R}$, support and confidence thresholds σ and δ , the CFD discovery problem is to identify a minimal cover Σ consisting of all minimal CFDs over r that satisfy the thresholds σ and δ .

4 SOLUTION OVERVIEW

As shown in Figure 1, SCFDM takes a relation r as input and outputs a minimal cover of CFDs. Each module operates as follows.

Relational encoding and attribute embedding. This module converts relation r into vector representations through a two-stage process (Section 5). It first maps discrete values to a zero-one matrix M , which constructs the high-dimensional Transformer input tensor $\mathcal{T}$ and supports representative tuple sampling. It then projects vectors in M into dense latent states E_i in $\mathcal{T}$, enabling dot-product self-attention to capture attribute-wise correlations in r . Meanwhile, row-wise similarity in M guides representative sampling, as similar tuples are more likely to support the same CFDs. This embedding strategy is well suited to CFD discovery. Each row of M encodes a tuple as concatenated zero-one vectors: matching one positions indicate equal attribute values ($=$), while different one positions indicate unequal values ($\neq$). Thus, M preserves the equality and inequality relationships required for CFD evaluation. The resulting tensor $\mathcal{T}$ provides structured input for the Transformer to learn attribute correlations, whose self-attention layers capture cross-attribute dependencies and generate informative patterns for efficient CFD discovery.

Correlated attribute sets discovery. This module models a CFD as a conditional probability distribution in order to identify correlated attribute sets (Section 6). For each consequent attribute $Y \in \text{attr}(\mathcal{R})$, it defines the correlated attribute set as $\text{Corr}_Y = \{Y\} \cup \{X \in \text{attr}(\mathcal{R}) \setminus \{Y\} \mid f_{\text{att}}(X, Y) \geq \theta_{\text{att}}\}$, where $f_{\text{att}}(\cdot)$ is the correlation

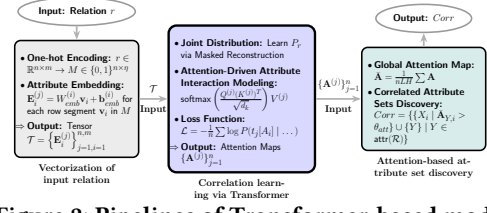


Figure 2: Pipelines of Transformer-based model

extraction function and θ_{att} is the structural significance threshold defined in Section 6.3. The correlated attribute sets are collected as $\text{Corr} = \{\text{Corr}_Y \mid Y \in \text{attr}(\mathcal{R})\}$. A Transformer model learns these distributions through masked value reconstruction [22, 45], capturing attribute correlations via self-attention. Based on the global attention maps [87], the module identifies candidate antecedent sets X for each consequent Y with quadratic complexity. These correlated attribute sets accelerate CFD discovery via data parallelism by guiding the partition of the relation into independent sub-tables.

Representative tuple sampling. This module extracts representative tuples from r with a discovery recall guarantee (Section 7). It first derives a sampling lower bound N , which reduces sample size by incorporating binomial variance. To preserve discovery recall, this module employs similarity-aware stratified sampling via locality sensitive hashing (LSH), partitioning tuples into similarity buckets in linear time. By adopting a diversity-first selection logic, it ensures uniform coverage across all data buckets and prevents low-support CFDs from being obscured by dominant patterns, thereby maintaining the diversity of CFDs within the sampling budget N .

Parallel discovery of CFDs. SCFDM projects the sampled representative tuples to multiple correlated attribute sets and produces sub-table inputs for CFD discovery. Then, SCFDM employs existing discovery algorithms to find CFDs from these independent sub-tables in parallel. That is, SCFDM can parallelize any CFD discovery algorithm via data parallelism (Section 8).

Specifically, Figure 2 illustrates the pipeline of the Transformer-based model in SCFDM that takes relation r as input and yields its correlated attribute sets Corr . This workflow encompasses relation vectorization (Section 5), correlation learning via Transformer (Section 6.2), and attention-based attribute set discovery (Section 6.3). The process begins with vectorization, which first applies one-hot encoding to relation r over $\text{attr}(\mathcal{R})$ to produce a zero-one matrix $M \in \{0, 1\}^{n \times \eta}$, where $\eta = \sum_{i=1}^m c_i$ is the domain size of the m attributes (each attribute A_i having $|\text{dom}(A_i)| = c_i$). An embedding layer then projects M into the high-dimensional tensor $\mathcal{T} \in \mathbb{R}^{n \times m \times d}$. This tensor $\mathcal{T}$ inputs to the learning module, where a Transformer encoder performs masked attribute modeling. By minimizing reconstruction loss $\mathcal{L}$ across all tuples, the model learns the conditional probability distribution $P_r(A_i \mid \text{attr}(\mathcal{R}) \setminus \{A_i\})$ and generates attention maps $\{A^{(j)}\}$ capturing attribute correlations. Finally, the module performs global analysis on these maps to construct a global map $\bar{A}$, which is iteratively leveraged to identify correlated antecedents X exceeding a significance threshold for each attribute Y , yielding Corr in $O(m^2)$ time.

5 VECTORIZATION OF INPUT RELATION

This section presents a two-stage vectorization of relation r , where one-hot encoding produces a binary matrix M (Section 5.1) and an attribute embedding layer projects M into tensor $\mathcal{T}$ (Section 5.2). This representation enables the Transformer to model attribute

correlations in CFDs via dot-product similarities in $\mathcal{T}$.

5.1 One-hot Encoding of Tuples

Given a relation r over the attribute set $\text{attr}(\mathcal{R}) = \{A_1, \dots, A_m\}$, let $\text{dom}(A_i)$ denote the domain of each attribute A_i with $c_i = |\text{dom}(A_i)|$. Each value $a \in \text{dom}(A_i)$ is encoded as a zero-one vector $\mathbf{v}_i \in \{0, 1\}^{c_i}$, where only the j -th element x_j is set as 1 if a is the j -th value (alphabetically ordered) in $\text{dom}(A_i)$. The representation of each tuple $t_j \in r$ is the concatenation of zero-one vectors $\mathbf{m}_j = [\mathbf{v}_1^\top, \mathbf{v}_2^\top, \dots, \mathbf{v}_m^\top]$ that corresponds to its m attribute values (ordered alphabetically by attribute names in $\text{attr}(\mathcal{R})$), where $\mathbf{m}_j \in \{0, 1\}^{1 \times \eta}$ and $\eta = \sum_{i=1}^m c_i$. Thus, the relation r is transformed into a zero-one valued matrix $M \in \{0, 1\}^{n \times \eta}$ by encoding each tuple as a zero-one vector.

5.2 Embedding of Attributes and Relations

Given the zero-one encoding matrix M (Section 5.1), we design an attribute embedding layer (Figure 2) for Transformer-based attribute correlation modeling [86]. Specifically, for a row $\mathbf{m}_j = [\mathbf{v}_1^\top, \mathbf{v}_2^\top, \dots, \mathbf{v}_m^\top]$ in M , each segment $\mathbf{v}_i \in \{0, 1\}^{c_i}$ ($1 \leq i \leq m$) is projected to a d -dimensional vector via an embedding layer (similar to [46]): $\mathbf{E}_i^{(j)} = W_{emb}^{(i)} \mathbf{v}_i + \mathbf{b}_{emb}^{(i)}$, where $\mathbf{E}_i^{(j)} \in \mathbb{R}^d$ represents the embedding of attribute A_i in tuple t_j . Thus, a tuple $t_j \in r$ is transformed into a matrix $\mathbf{H}_{in}^{(j)} = [\mathbf{E}_1^{(j)}, \dots, \mathbf{E}_m^{(j)}]^\top \in \mathbb{R}^{m \times d}$ and we denote the trainable weights of the attribute embedding layer as $\Theta_{emb} = \{W_{emb}^{(i)}, \mathbf{b}_{emb}^{(i)}\}_{i=1}^m$ (weight details refer to Section 6.2 and Appendix D in [1]). These parameters Θ_{emb} are randomly initialized and optimized with the downstream Transformer (Section 6). By embedding all n rows in M , the attribute embedding layer produces a tensor $\mathcal{T} \in \mathbb{R}^{n \times m \times d}$, which enables the Transformer to treat the relation as a sequence of m attribute embeddings for each of the n tuples. This vectorized sequence enables the Transformer to capture attribute correlations via self-attention.

Remark. Modeling relation r as dense embeddings $\mathbf{E}_i$ enables the discovery of attribute correlations. Self-attention in a uniform d -dimensional space computes dot-product similarities between attributes, making hidden CFD correlations measurable via embeddings and quantifying the relationship between a candidate antecedent set $\mathbf{X}$ and a consequent Y .

6 CORRELATED ATTRIBUTE DISCOVERY

In this section, we first formulate CFDs as conditional probability distributions (Section 6.1). Based on this formulation, we train a Transformer model using the embedding matrix $\mathcal{T}$ derived in Section 5.2 to learn the attribute correlations by modeling these conditional distributions (Section 6.2). Finally, we demonstrate how the attention maps within the trained Transformer can be utilized to capture and identify correlated attribute sets in CFDs (Section 6.3).

6.1 CFD as Conditional Probability

Inspired by the statistical analysis of FDs in [92], we first analyze CFDs from a statistical perspective. Following [36, 92], consider a tuple set $r_\varphi \subseteq r$, where $r_\varphi = \{t \in r \mid t[X] \leq t_p[X] \wedge t[Y] \leq t_p[Y]\}$, and for any two tuples $t_i, t_j \in r_\varphi$ ($i \neq j$), if $t_i[X] = t_j[X]$, then $t_i[Y] = t_j[Y] \leq t_p[Y]$. A CFD $\varphi : (\mathbf{X} \rightarrow Y, t_p)$ is a statement over the attribute set $\mathbf{X} \subseteq \text{attr}(\mathcal{R})$ and an attribute $Y \in \text{attr}(\mathcal{R})$, where

Algorithm 1 AttrFinder (Attention-based Discovery)

Input: Relation r , transaction matrix $M \in \{0, 1\}^{n \times \eta}$, mask probability p_{mask} , scaling factor z , attention threshold θ_{att}
Output: Correlated attribute sets $Corr$
 $Corr \leftarrow \emptyset$
while not converged **do**
 $\mathcal{B} \leftarrow \text{SampleBatch}(M)$
 $\tilde{\mathcal{B}} \leftarrow \text{ApplyMask}(\mathcal{B}, p_{mask})$
 $\mathcal{T}_{\tilde{\mathcal{B}}} \leftarrow \text{Embedding}(\tilde{\mathcal{B}})$
 $\hat{Y} \leftarrow \text{TransformerEncoder}(\mathcal{T}_{\tilde{\mathcal{B}}})$
 $\mathcal{L} \leftarrow \frac{1}{|\tilde{\mathcal{B}}|} \sum_{j \in \tilde{\mathcal{B}}} \text{CrossEntropy}(M_j, \hat{Y}_j)$
 UpdateWeights($\mathcal{L}$)
 $\hat{\mathbf{A}} \leftarrow \frac{1}{n \cdot L \cdot H} \sum_{j=1}^n \sum_{l=1}^L \sum_{h=1}^H \mathbf{A}_{j,l,h}$
 for each attribute $Y \in \text{attr}(\mathcal{R})$ **do**
 $\text{Row}_Y \leftarrow \hat{\mathbf{A}}[Y, \cdot]$
 $\theta_{att} \leftarrow z/m$
 $\mathbf{X}_Y \leftarrow \{A_i \mid \hat{\mathbf{A}}[Y, i] \geq \theta_{att}, A_i \neq Y\}$
 if $\mathbf{X}_Y$ is not empty **then**
 $Corr_Y \leftarrow \mathbf{X}_Y \cup \{Y\}$
 $Corr \leftarrow Corr \cup \{Corr_Y\}$
return $Corr$

an assignment to $\mathbf{X}$ uniquely determines the value of Y on the subset r_φ above support and confidence thresholds. Thus, $\wedge_{A \in \mathbf{X}} t_i[A] = t_j[A]$ and $t_i[Y] = t_j[Y]$ can be viewed as two associated events, and the conditional probability $P_r(t_i[Y] = t_j[Y] \mid \wedge_{A \in \mathbf{X}} t_i[A] = t_j[A])$ captures the dependency between $\mathbf{X}$ and Y . φ holds with confidence δ when this conditional probability is at least δ .

Given a CFD $\varphi : (\mathbf{X} \rightarrow Y, t_p)$ and a tuple set $r_\varphi \subseteq r$ that conforms with φ , φ holds with confidence δ on r_φ if the conditional probability $P_r(t_i[Y] = t_j[Y] \mid \wedge_{A \in \mathbf{X}} t_i[A] = t_j[A])$ is at least δ for any tuples $t_i, t_j \in r_\varphi$. Here, $P_r(t_i[Y] = t_j[Y] \mid \wedge_{A \in \mathbf{X}} t_i[A] = t_j[A]) \geq \delta$ is a threshold condition on the conditional probability between $\mathbf{X}$ and Y over r_φ , not a conditional probability itself. Consider the CFD φ in Definition 3.3, where $\mathbf{X} = \{A_1, A_2, \dots, A_k\}$ is a multi-variable set and Y is a variable. The domain of $\mathbf{X}$, denoted as $\text{dom}(\mathbf{X})$, is determined by r_φ , i.e., $\text{dom}(\mathbf{X}) = \{t_i(A_1) \times t_i(A_2) \times \dots \times t_i(A_k)\}$ for all $t_i \in r_\varphi$. When $\mathbf{X}$ is assigned by values in $\text{dom}(\mathbf{X})$, there exists a function of conditional association $Y = f(\mathbf{X})$ from $\mathbf{X}$ to Y such that for any $\mathbf{x} \in \text{dom}(\mathbf{X})$ with $y = f(\mathbf{x})$, the conditional probability $P_r(Y = y \mid \mathbf{X} = \mathbf{x})$ is at least δ . That is, when $\mathbf{X}$ takes values in its domain $\text{dom}(\mathbf{X})$, the conditional probability that $Y = f(\mathbf{X})$ reaches the confidence threshold δ , as required by Definition 3.3.

6.2 Correlation Learning via Transformer

In order to learn the conditional probability of CFDs, we formulate an end-to-end reconstruction task for the Transformer, where correlations among attributes are captured by learning the conditional probability $P_r(t_i[Y] = t_j[Y] \mid \wedge_{A \in \mathbf{X}} t_i[A] = t_j[A])$. Specifically, we treat each tuple vector in $\mathcal{T}$ as a sequence and train the Transformer model to recover masked attribute values, thereby completing the learning of conditional probabilities in CFDs driven by this representation reconstruction task.

For each tuple t_j , each attribute $A_i \in \text{attr}(\mathcal{R})$ is iteratively masked with a [MASK] token. An L -layer Transformer applies self-attention over the remaining $m - 1$ attributes to obtain a hidden representation that captures attribute correlations:

$$\text{Attention}(Q^{(j)}, K^{(j)}, V^{(j)}) = \text{softmax}\left(\frac{Q^{(j)}(K^{(j)})^T}{\sqrt{d_k}}\right)V^{(j)}, \text{ where}$$

$Q^{(j)}, K^{(j)}, V^{(j)}$ are linear projections of the attribute embeddings $\mathbf{H}_{in}^{(j)}$ in $\mathcal{T}$. Notably, $\mathbf{A}^{(j,l,h)} = \text{softmax}(Q^{(j,l,h)}(K^{(j,l,h)})^T / \sqrt{d_k})$ is the self-attention map that quantifies attribute dependencies. **Stacking L layers ($L = 6$ as commonly used for tabular data [42, 45]) enables iterative refinement of attribute dependencies, e.g., capturing cases where X_1 and X_2 jointly determine Y . To capture dependencies among multiple attributes, the multi-head self-attention uses four attention heads, whose attention maps are averaged in Section 6.3.**

The attribute embedding layer and the Transformer model are optimized by minimizing the average cross-entropy loss across all tuples: $\mathcal{L}(\Theta) = -\frac{1}{n} \sum_{j=1}^n \log P(t_j[A_i] \mid t_j[\text{attr}(\mathcal{R}) \setminus \{A_i\}]; \Theta)$, which forces the Transformer to learn the conditional probability distribution P_r of relation r , where $\Theta = \{\Theta_{\text{emb}}, \Theta_{\text{trans}}\}$ is iteratively optimized including the attribute embedding parameters Θ_{emb} and the Transformer parameters Θ_{trans} . By minimizing $\mathcal{L}(\Theta)$, the Transformer maximizes the likelihood of predicting the true attribute values, thereby aligning its internal attention mechanism with the conditional distributions P_r of the CFDs. This optimization compels the Transformer to discover attribute correlations by focusing on attributes that provide the relatively stronger predictive signals for the masked targets. The parameters $\Theta = \{\Theta_{\text{emb}}, \Theta_{\text{trans}}\}$ are adjusted through backpropagation using the AdamW optimizer [60]. The optimization and training phase are executed on two 32GB NVIDIA GPUs. Training details refer to Appendix D in [1].

6.3 Attention-based Attribute Set Discovery

The extraction of correlated attribute sets in CFDs is achieved through the analysis of attention maps in the trained Transformer of Section 6.2. As shown in Algorithm 1, AttrFinder takes two steps to identify correlated attribute sets.

Step 1: Attention map extraction (Lines 4–11). Following Section 6.2 (Lines 4–10), this optimization process compels the attention maps $\{\mathbf{A}^{(j)}\}_{j \in \mathcal{B}}$ to quantify the attribute correlations required to model the conditional probabilities of CFDs. Then, AttrFinder computes the global attention map $\bar{\mathbf{A}} \in \mathbb{R}^{m \times m}$ by averaging the attention maps across all n tuples, L layers, and H heads (Line 11): $\bar{\mathbf{A}} = \frac{1}{n \cdot L \cdot H} \sum_{j=1}^n \sum_{l=1}^L \sum_{h=1}^H \mathbf{A}^{(j,l,h)}$, where each entry $\bar{A}_{i,k}$ quantifies the correlation between attribute A_i and A_k . A higher value of $\bar{A}_{i,k}$ indicates a stronger conditional probability P_r in CFDs.

Step 2: Correlated attribute sets identification (Lines 12–19). AttrFinder utilizes the global attention map $\bar{\mathbf{A}}$ in Step 1 to identify correlated attribute sets of CFDs. By iteratively treating each attribute $A_i \in \text{attr}(\mathcal{R})$ as a consequent Y (Line 12), AttrFinder retrieves the row corresponding to attribute Y from the global attention map $\bar{\mathbf{A}}$ (Line 13), which captures the cumulative influence of all other attributes on Y . We then determine the candidate antecedent set X by selecting attributes whose value of $\bar{A}_{i,k}$ exceed a threshold θ_{att} . Following common practice in attention analysis [21, 87], we set $\theta_{\text{att}} = \frac{z}{m}$ (Line 14), where $z > 1$ is a scaling factor, filtering out attributes with negligible predictive contribution relative to a uniform distribution of $1/m$. For each consequent attribute Y , the resulting correlated attribute set is $\text{Corr}_Y = \{X_i \mid \bar{A}_{Y,i} > \theta_{\text{att}}\} \cup \{Y\}$ (Lines 15–17). The collection of correlated attribute sets for relation r is obtained by iterating over all attributes in $\text{attr}(\mathcal{R})$: $\text{Corr} = \{\text{Corr}_Y \mid Y \in \text{attr}(\mathcal{R})\}$ (Line 18).

Algorithm 2 RepSampler (Similarity-based Stratified Sampling)

```

1: Input: Relation  $r$ , thresholds  $\sigma, \delta, \epsilon$ , Chernoff coefficient  $\tau$ , LSH parameters  $L$ , zero-one encoding matrix  $M$ 
2: Output: Representative sample set  $S$ 
3:  $N \leftarrow$  calculate sampling lower bound via Theorem 2
4:  $\mathcal{B} = \{B_1, \dots, B_k\} \leftarrow$  Partition  $M$  via LSH  $h(\cdot)$  using  $L$ 
5:  $S \leftarrow \emptyset, \mathcal{B} \leftarrow$  sort buckets in  $\mathcal{B}$  in ascending order of size  $|B_i|$ 
6: while  $|S| < N$  and  $\mathcal{B}$  is nonempty do
7:   for each bucket  $B_i \in \mathcal{B}$  do
8:     if  $B_i$  is nonempty then
9:        $s \leftarrow$  randomly sample a tuple from  $B_i$ 
10:       $S \leftarrow S \cup \{s\}, B_i \leftarrow B_i \setminus \{s\}$ 
11:      if  $|S| = N$  then
12:        break
13:      else
14:         $\mathcal{B} \leftarrow \mathcal{B} \setminus \{B_i\}$ 
15: return  $S$ 
```

Example 4: We provide a walkthrough of AttrFinder using Table 1, which discovers the correlated attribute set for the consequent $Y = \text{City}$, conforming to the conditional probability formulation in Section 6.1. After optimization, the weights in the global attention map $\bar{\mathbf{A}}$ reflect the learned conditional distributions P_r . For the row of *City*, suppose the weights are $\bar{\mathbf{A}}[\text{City}, \text{Gender}] = 0.35$, $\bar{\mathbf{A}}[\text{City}, \text{Proto}] = 0.42$, and $\bar{\mathbf{A}}[\text{City}, \text{Age}] = 0.04$.

Given $m = 14$ and $z = 3$, the threshold is $\theta_{\text{att}} = 3/14 \approx 0.21$. According to Step 2, as the scores for *Gender* and *Proto* exceed θ_{att} , they indicate a strong conditional dependency with respect to *City*, whereas *Age* ($0.04 < 0.21$) falls below the threshold and is pruned. This extracts the correlated attribute set $\text{Corr}_{\text{City}} = \{\text{Gender}, \text{Proto}, \text{City}\}$, establishing the context required for sub-table partitioning.

Complexity. AttrFinder’s complexity is $O(n \cdot m^2)$ for n tuples and m attributes; other factors (epochs, layers, and embedding dimension) are treated as constants (see Appendix E in [1] for details).

7 SAMPLING WITH ACCURACY BOUNDS

Mining CFDs on large datasets remains costly as the number of tuples grows. To avoid rule loss caused by improper sampling [31], we propose RepSampler, an accuracy-guaranteed sampling algorithm for efficient CFD discovery. We first derive a sampling lower bound (Section 7.1), and then select representative tuples from stratified partitions to preserve CFD diversity (Section 7.2).

7.1 Lower Bound for Sampling

To mitigate CFD loss during tuple sampling, we investigate the lower bound of the sample size with accuracy guarantees. Let Σ be the ground-truth CFDs holding on the original relation r , and Σ_s be the CFDs discovered from the sampled relation r_s . We evaluate discovery quality via precision and recall: $\text{precision}(\Sigma, \Sigma_s) = \frac{|\Sigma_s \cap \Sigma|}{|\Sigma_s|}$ reports the proportion of discovered CFDs that are valid on r , while $\text{recall}(\Sigma, \Sigma_s) = \frac{|\Sigma_s \cap \Sigma|}{|\Sigma|}$ measures the fraction of ground-truth CFDs recovered from r_s . Intuitively, precision quantifies the percentage of spurious rules generated by sampling, whereas recall quantifies the loss of CFDs incurred by the sampling process.

Given a CFD φ , and N independently and identically distributed samples $t_1, t_2, \dots, t_N$ randomly drawn from the relation r , we define $X_i (1 \leq i \leq N)$ as an indicator variable where $X_i = 1$ if t_i conforms with φ and $X_i = 0$ otherwise. The sum $X = \sum_{i=1}^N X_i$ follows a binomial distribution $X \sim B(N, \hat{\sigma})$, with $E[X] = N\hat{\sigma}$ and $\text{Var}(X) = N\hat{\sigma}(1 - \hat{\sigma})$, where $\hat{\sigma} = \sigma/n$ is the normalized support. Let

ϵ be the additive estimation error [6, 23] between the estimated and true normalized support, satisfying $Pr(|\hat{\sigma}_s - \hat{\sigma}| \leq \epsilon) \geq 1 - \delta\beta$, where $\hat{\sigma}_s$ and $\hat{\sigma}$ denote the estimated and true normalized support, respectively. To derive a sampling lower bound N under recall guarantee β and error bound ϵ , we consider the following concentration inequalities. Notably, since X is a sum of Bernoulli trials, we can utilize the variance-sensitive form of concentration bounds (Bernstein-type) to obtain a tighter N compared to the standard Hoeffding's inequality [44]. First, we apply the variance-augmented Hoeffding-Bernstein inequality [11, 44]: $Pr(|X - E[X]| \geq N\epsilon) \leq 2 \exp\left(\frac{-(N\epsilon)^2}{2\text{Var}(X) + \frac{2}{3}N\epsilon}\right) \approx$

$2 \exp\left(\frac{-2(N\epsilon)^2}{\text{Var}(X)}\right) \leq 1 - \delta\beta$. The use of $\text{Var}(X)$ is justified since it accounts for the distribution's sparsity; when $\hat{\sigma}$ is small (typical for CFD discovery), $\text{Var}(X) \ll N$, yielding a tighter bound than the non-parametric version. From this inequality, we can derive the variance-augmented lower bound: $N \geq \frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{1-\delta\beta}{\epsilon})}{-2\epsilon^2}$.

We next compare the proposed variance-augmented lower bound with commonly used bounds derived from other sampling inequalities [6, 17, 23]. Specifically, using Chebyshev's inequality $Pr(|X - E[X]| \geq N\epsilon) \leq \frac{\text{Var}(X)}{(N\epsilon)^2} \leq 1 - \delta\beta$ [23], we obtain Chebyshev's lower bound $N \geq \frac{\hat{\sigma}(1-\hat{\sigma})}{\epsilon^2(1-\delta\beta)}$. According to Bennett's inequality $Pr(|X - N\hat{\sigma}| \geq N\epsilon) \leq 2 \exp(-N\hat{\sigma}\Theta(\frac{\epsilon}{\hat{\sigma}})) \leq 1 - \delta\beta$ [6], we can derive Bennett's lower bound $N \geq \frac{-\ln(1-\delta\beta)}{(\hat{\sigma}+\epsilon) \ln(1+\frac{\epsilon}{\hat{\sigma}}) - \epsilon}$. Finally, a hybrid Chernoff bound was introduced in [17], given by $N \geq \frac{3\tau}{\epsilon^2} \ln\left(\frac{2}{1-\delta\beta}\right)$.

Theorem 1: For $X \sim B(N, \hat{\sigma})$ where $0 \leq \hat{\sigma} \leq 1$, the variance-augmented lower bound is tighter than both Chebyshev's and Bennett's lower bounds.

Proof. Please refer to Appendix C.1 of [1] for the proof. $\square$

Theorem 2: Given the normalized support threshold $\hat{\sigma}$, additive Chernoff error bound ϵ , error ratio τ , confidence δ , and recall β , when $X \sim B(N, \hat{\sigma})$ and $\hat{\sigma}(1 - \hat{\sigma}) \leq 6\tau$, the sampling lower bound N is:

$$N = \begin{cases} \frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{2}{1-\delta\beta})}{2\epsilon^2}, & \hat{\sigma} \in [0, \hat{\sigma}_1] \cup (\hat{\sigma}_2, 1] \text{ when } \tau \leq \frac{1}{24} \\ \frac{3\tau}{\epsilon^2} \ln\left(\frac{2}{1-\delta\beta}\right), & \hat{\sigma} \in (\hat{\sigma}_1, \hat{\sigma}_2] \text{ when } \tau \leq \frac{1}{24} \\ \frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{2}{1-\delta\beta})}{2\epsilon^2}, & \hat{\sigma} \in [0, 1] \text{ when } \tau > \frac{1}{24} \end{cases}$$

where $\hat{\sigma}_1, \hat{\sigma}_2 = \frac{1 \pm \sqrt{1-24\tau}}{2}$.

Proof. Please see Appendix C.2 of [1] for the proof. $\square$

Example 5: The following example illustrates how to derive the sample size bound in Theorem 2. Consider the parameters $\hat{\sigma} = 0.01$, $\epsilon = 0.001$, $\delta = 0.999$, $\tau = 0.02$, and $\beta = 1$. AttrFinder first computes the threshold $\hat{\sigma}_1 = \frac{1-\sqrt{1-24 \times 0.02}}{2} \approx 0.142$. Since the true normalized support satisfies $\hat{\sigma} = 0.01 \leq \hat{\sigma}_1$ for $\tau \leq \frac{1}{24}$, the required sample size is $N = \frac{0.01 \times 0.99 \times \ln(2000)}{2 \times 0.001^2} \approx 37,624$. In contrast, the hybrid Chernoff bound [17] requires $N = \frac{3 \times 0.02}{0.001^2} \ln(2000) \approx 456,055$. Thus, the proposed bound reduces the required sample size by more than 91.7%.

7.2 Representative Tuple Sampling

Existing sampling methods often overlook low-support CFDs. We address this by stratifying the zero-one matrix M from Section 5.1 into buckets, ensuring low-support CFDs are represented. We then sample tuples from each bucket under the bound N in Section 7.1 to

Algorithm 3 PCFDFinder (Data-parallel CFD mining)

```

1: Input: Correlated attribute sets  $Corr = \{Corr_1, Corr_2, \dots, Corr_l\}$ ,
   the representative tuple set  $S$ , a sequential CFD mining algorithm  $\Lambda$ ,
   support  $\sigma$ , confidence  $\delta$ , and the relation  $r$ 
2: Output: A minimal CFD cover  $\Sigma_s$  satisfying thresholds
3:  $R_{sub} = \{r_1, r_2, \dots, r_l\} \leftarrow$  Generate from  $Corr$  and  $S$ 
4: Create  $l$  threads across multiple processors
5: for each thread  $t_i$  for  $i \in [1, l]$  executing in parallel do
6:    $\Sigma_{si} \leftarrow$  Mine the cover of minimal CFDs from  $r_i$  using  $\Lambda$ 
7:   for each CFD  $\phi$  in  $\Sigma_{si}$  do
8:     Verify the CFD  $\phi$  on  $r$  and remove rules from  $\Sigma_{si}$  that violate  $r$ 
9:  $\Sigma_s \leftarrow \Sigma_{s1} \cup \Sigma_{s2} \cup \dots \cup \Sigma_{sl}$ 
10: return  $\Sigma_s$ 

```

construct S , preserving the sampling guarantees and CFD diversity.

Similarity-based Representative Sampling. Since each row $m_i \in M$ concatenates zero-one vectors, the inner product $m_i \cdot m_j$ counts the shared attribute values between tuples t_i and t_j . To capture both high- and low-support CFDs, we propose similarity-aware stratified sampling, which uses locality-sensitive hashing (LSH) [49] to group rows of M into collision buckets by their similarity, enabling representative sampling from r .

As shown in Algorithm 2, RepSampler samples representative tuples to preserve diverse CFDs. For the zero-one matrix M , it uses MinHash-based LSH [12] to map tuples into collision buckets $\mathcal{B} = \{B_1, \dots, B_k\}$ (Line 4). MinHash processes only non-zero indices, avoiding computation over zero entries and generating each row's signature in a single scan. Tuples in the same bucket have high Jaccard similarity [12] with high probability and are therefore likely to share the same CFDs, with an $O(n)$ time complexity [49]. Unlike random sampling, which may miss low-support CFDs [74, 85], or frequency-/value-based stratification, which incurs high overhead in high-dimensional spaces [5, 59], LSH efficiently groups similar tuples and preserves rare patterns with low overhead. RepSampler first removes buckets B_i with $|B_i| < \sigma$, since they cannot support a CFD reaching the support threshold. It then processes the remaining buckets in ascending order of size, sampling one tuple from each non-empty bucket per round until $|S|$ reaches the bound N (Lines 5–15). This prioritizes small buckets, helping preserve low-support CFD patterns that may be overlooked by large, high-support buckets. When more than N non-empty buckets remain ($k > N$), the N smallest buckets are processed first, thereby yielding $|S| = N$ in the first iteration (Line 11), after which the loop terminates. The sample size is bounded by N from Theorem 2 (Line 3).

Example 6: We illustrate the stratified sampling workflow in Algorithm 2 using Table 1. Suppose LSH partitions the relation into buckets $\mathcal{B} = \{B_1 = \{t_5\}, B_2 = \{t_3, t_4\}, B_3 = \{t_1, t_2, t_6, t_7\}\}$ (Line 4), and the derived lower bound is $N = 5$ (Line 3). To prevent pruning of low-support patterns, RepSampler iteratively rotates through the buckets in $\mathcal{B}$ to sample tuples: (1) Iteration 1 (Lines 6–14): It randomly extracts t_5 from B_1 , t_3 from B_2 , and t_2 from B_3 , yielding $S = \{t_5, t_3, t_2\}$. At this point, B_1 becomes empty and is pruned (Line 13). (2) Iteration 2: Since $|S| = 3 < N$, it continues sampling from the remaining non-empty buckets, randomly selecting t_4 from B_2 and t_7 from B_3 . The sample size now reaches $|S| = 5 = N$, triggering the break condition (Line 11). The final representative set $S = \{t_5, t_3, t_2, t_4, t_7\}$ protects the rare bucket B_1 , preserving the low-support pattern.

Complexity. The time complexity of RepSampler is linear in the

tuple count, i.e., $O(n)$, treating attributes and hash functions as constants (see Appendix G in [1] for details).

8 PARALLEL RULE DISCOVERY

We propose PCFDFinder, a CFD discovery framework that leverages correlated attribute sets (Section 6) and representative tuples (Section 7) to enable parallel execution of existing CFD mining algorithms. We first present its sub-table-based parallel design (Section 8.1), and then analyze its computational efficiency (Section 8.2).

8.1 Parallel Discovery Framework

To facilitate parallel discovery, we decompose the representative tuple set S (obtained in Section 7.2) into l sub-tables $\{r_{s1}, \dots, r_{sl}\}$. This decomposition is guided by the correlated attribute sets $Corr = \{corr_1, \dots, corr_l\}$ derived in Section 6.3, where each sub-table r_{si} is generated by projecting S onto the corresponding attribute set $corr_i$ (i.e., $r_{si} = \pi_{corr_i}(S)$). We then define cover $\Sigma_s = \bigcup \Sigma_{si}$, where each cover Σ_{si} is the set of minimal CFDs discovered from the corresponding sub-table r_{si} .

Next, we introduce PCFDFinder, a parallel framework for mining CFDs. As outlined in Algorithm 3, PCFDFinder aims to discover the minimal cover Σ_s of CFDs from r in parallel (line 2). PCFDFinder projects the representative tuple set S onto correlated attribute sets $Corr = \{corr_1, corr_2, \dots, corr_l\}$, and obtains l input sub-tables $R_{sub} = \{r_{s1}, \dots, r_{sl}\}$ (Line 3). Then, PCFDFinder spawns l threads to concurrently run a sequential CFD discovery algorithm Λ on each sub-table r_{si} (Lines 4-6). Within each thread, every discovered candidate CFD ϕ is immediately validated against the original instance r with a complexity of $O(n)$, where n is the number of tuples in r , and CFDs violating r are discarded from local covers (Lines 7-8). *This verification checks whether ϕ satisfies the required support and confidence thresholds on r (support $\geq \sigma$ and confidence $\geq \delta$). However, satisfying these thresholds alone does not guarantee that ϕ belongs to the ground-truth cover Σ . The ground truth Σ is obtained through exhaustive mining and expert verification (Section 9) and is used to compute accuracy.* Once all threads complete, PCFDFinder computes the union of the validated local covers: $\Sigma_s = \bigcup_{i=1}^l \Sigma_{si}$ (Line 9). The final cover Σ_s is returned as the cover of CFDs on relation r (Line 10).

8.2 The reduction of computation

The time complexity of the sequential CFD mining algorithm is $O(n \times m \times d^m)$ [73], where n , m and d represent the number of tuples, attributes, and the average domain size of attributes, respectively. With data parallelism in PCFDFinder, this time complexity is reduced to $O(n_i \times m_i \times d_i^{m_i})$, where n_i , m_i , and d_i are the number of tuples, attributes, and domain size in the largest sub-table. As a result, CFD discovery efficiency is improved by a factor of $\theta = O(\frac{n \times m \times d^m}{n_i \times m_i \times d_i^{m_i}}) \geq O(\frac{d^{(1-\kappa_2) \times m}}{\kappa_1 \times \kappa_2})$, where $d_i \leq d$, and κ_1, κ_2 denote the ratios of tuple counts and attribute counts in the largest sub-table relative to the original relation, respectively. In parallel execution, PCFDFinder delivers linear speedup via the tuple scaling factor κ_1^{-1} and exponential efficiency gains via the attribute reduction factor κ_2^{-1} across decomposed sub-tables.

9 EXPERIMENTAL STUDY

In this section, we experimentally evaluated (1) the efficiency of SCFDM; (2) the scalability of SCFDM; (3) the effectiveness (discovery accuracy) of SCFDM; (4) sensitivity to CFD core parameters; (5) the ablation study of SCFDM; and (6) a case study of CFDs on real-world and synthetic datasets.

9.1 Experimental Setting

Datasets. As shown in Table 2 (details in Appendix H.9 of [1]), we evaluate SCFDM on five real-world datasets (1–5) from UCI [3] and Kaggle [71], and two synthetic datasets (6–7) generated via Bayesian networks [79, 89, 92]: (1) RT-IOT (RT), covering IoT network traffic and cyberattacks; (2) Census-Income (Census), containing weighted 1994–1995 population survey data; (3) Crop, combining multi-modal remote sensing images; (4) Flight, recording domestic flight on-time performance; and (5) Adult, predicting census-based income levels. (6) Syn-1 and (7) Syn-2 provide controlled environments with ground-truth CFDs known by construction, with 5% dependency noise [14] injected into Syn-1 and 5% cluster noise [14] injected into Syn-2 for robustness evaluation. Generated from Bayesian networks, they preserve realistic attribute correlations and provide known dependency structures for evaluating AttrFinder. They also differ in the number of attributes, allowing us to evaluate SCFDM across different schema sizes. In Table 2, the “#Sampled” column reports the representative tuples selected by RepSampler, while the “#Sub-tables” column lists the partitioned sub-tables generated by SCFDM. Since AttrFinder extracts a correlated attribute set $Corr_Y$ for each attribute $Y \in attr(\mathcal{R})$, the number of sub-tables equals the number of attributes.

Algorithms. We implement eight CFD discovery algorithms in C++: (1) CFDMiner [30] for mining constant CFDs; (2) CTANE [30], a level-wise algorithm for mining both constant and variable CFDs; (3) Itemset-First [73] and (4) FD-First [73], two advanced hybrid CFD mining methods that alternate between itemset mining and FD discovery via depth-first or FD-first strategies; (5) SCFDM_{part}, our proposed constant CFD miner integrating AttrFinder, RepSampler, and PCFDFinder (with CFDMiner as the sequential engine Λ); (6) SCFDM_{all}, our proposed miner for both constant and variable CFDs, employing Itemset-First as Λ ; and (7) SCFDM_{row} and (8) SCFDM_{col}, two ablation variants of SCFDM_{part} that omit AttrFinder and RepSampler, respectively. For fair comparison, we parallelized all four sequential baselines (CFDMiner, CTANE, Itemset-First, and FD-First) using parallel lattice traversal and multi-space search strategies [31, 53, 78] described in Section 2.

Ground truth of CFDs. For real-world datasets, we employ a dual-mining strategy within Metanome [66]: the level-wise CTANE [30] for exhaustive CFD discovery, supplemented by CFDMiner [30] for constant CFDs. For synthetic datasets, the ground truth Σ is inherently defined by the Bayesian generative process, where structural priors ensure that instances strictly adhere to the predefined dependency logic. Following automated generation/mining, manual expert review filters out redundant or invalid rules. This rigorous workflow yields a high-fidelity “gold standard” Σ for each dataset, with the total counts recorded in the “#CFDs” column of Table 2.

Table 2: Summary of datasets and algorithm outputs

Dataset	Source	#Tuples	#Sampled	#Attributes	#CFDs	#Sub-tables
RT	UCI	123, 117	8, 519	85	50, 498	85
Census	UCI	199, 523	10, 544	45	84, 309	45
Crop	Kaggle	325, 835	16, 063	175	78, 232	175
Flight	Kaggle	1, 000, 000	41, 057	25	6, 224	25
Adult	UCI	47, 621	8, 422	14	435	14
Syn-1	bnlearn	20, 000	8, 408	48	2, 263	48
Syn-2	bnlearn	20, 000	8, 397	37	1, 755	37

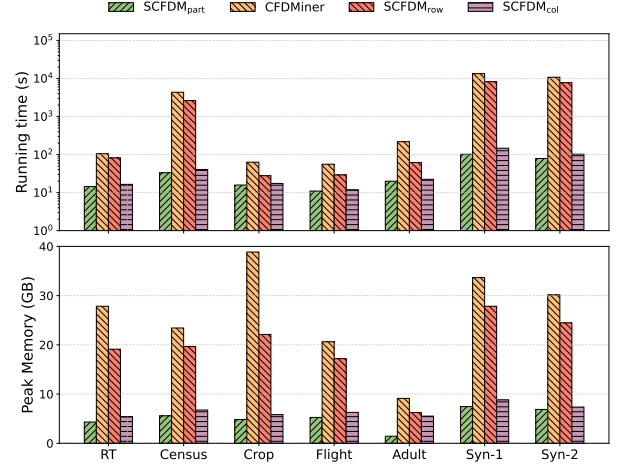
9.2 Experimental Results

Accuracy metrics. Let Σ denote the ground-truth CFDs of the original relation r , and Σ_s the CFDs discovered from the partitioned sub-tables r_s . Both the ground-truth and the sub-table mining use the same deterministic construction procedure. Therefore, differences between Σ and Σ_s stem from tuple sampling and attribute partitioning, not from alternative minimal covers. We evaluate SCFDM using the following metrics: (1) Precision (P) measures the proportion of discovered CFDs (Σ_s) that are valid on the original relation r : $P(\Sigma, \Sigma_s) = \frac{|\Sigma_s \cap \Sigma|}{|\Sigma_s|}$; (2) Recall (R) measures the fraction of ground-truth CFDs (Σ) successfully recovered in Σ_s : $R(\Sigma, \Sigma_s) = \frac{|\Sigma_s \cap \Sigma|}{|\Sigma|}$; (3) F1-score (F_1) is the harmonic mean of P and R : $F_1 = 2 \cdot \frac{P \cdot R}{P + R}$.

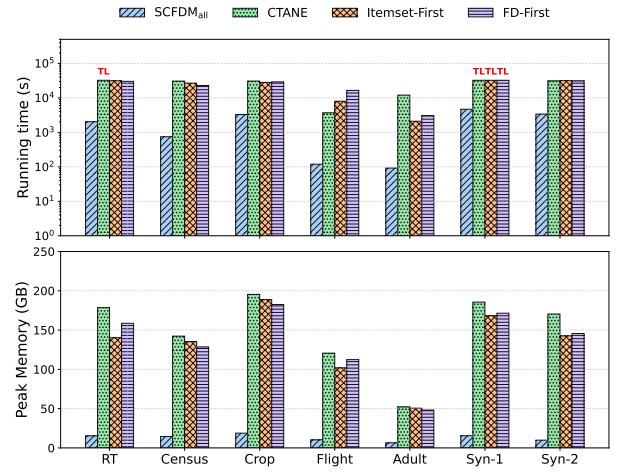
Default settings. Experiments run on a four-node cluster equipped with 224 GB RAM, where each node has two 28-core 2.70 GHz Intel Xeon CPUs. The Transformer training utilizes two NVIDIA GPUs (32 GB VRAM each) on one node. The embedding network uses a hidden dimension of 128 and is trained with a batch size of 128, optimized via AdamW with a learning rate of 5×10^{-4} for 20 epochs (patience = 3). Default parameters are: support $\sigma = 200$, confidence $\delta = 0.95$, support error bound $\epsilon = 5 \times 10^{-4}$, Chernoff ratio $\tau = 0.05$, and recall $\beta = 0.9$. The maximum CFD size k is set per consequent attribute via the entropy-driven strategy of Appendix H.3 of [1], yielding $k_Y \in [2, 7]$ in practice. Following [4, 43, 76], ϵ and τ are adaptively reduced for larger datasets (e.g., Crop and Flight) to better preserve low-support CFDs, which increases the sampled tuples in Table 2. The time limits are 4 hours for constant CFD discovery and 9 hours for both constant and variable CFDs. We report the average of three runs. We provide a guidance of hyperparameter configuration in Appendix H.4 of [1].

Exp-1: Efficiency of SCFDM. We evaluate the efficiency of SCFDM by comparing it with state-of-the-art baselines across all seven datasets. Figure 3 illustrates the execution time and peak memory usage, where “TL” denotes that a baseline exceeded the designated time limit (14, 400s for Figure 3(a) and 32, 400s for Figure 3(b)).

Figure 3(a) compares SCFDM_{part}, SCFDM_{row}, and SCFDM_{col} against CFDMiner. SCFDM_{part} consistently exhibits the highest efficiency, achieving an average speed-up of 105.18× across all datasets. Notably, on the Syn-2 dataset, CFDMiner took 10, 776.90s, whereas SCFDM_{part} completed in 78.96s, yielding a speedup of 136.49×. The peak memory curves in Figure 3(a) demonstrate that the runtime improvements are accompanied by reduced memory consumption. While CFDMiner suffers from combinatorial lattice explosion, yielding an average peak memory consumption of 26.23GB (up to 38.85GB on Crop), SCFDM_{part} reduces the average peak memory to 5.11GB, achieving a 5.13× reduction. Moreover, removing AttrFinder (SCFDM_{row}) increases average peak memory to 19.51GB due to full attribute traversal, while omitting RepSampler (SCFDM_{col}) leads to an average rise to 6.56GB due to additional tuple processing. Together, these components effectively reduce



(a) Runtime and memory usage comparison of SCFDM_{part} and baselines



(b) Runtime and memory usage comparison of SCFDM_{all} and baselines

Figure 3: Efficiency evaluation of SCFDM and baselines

memory consumption in CFD discovery.

Figure 3(b) evaluates SCFDM_{all} against CTANE, Itemset-First, and FD-First. SCFDM_{all} maintains a substantial lead, especially on complex datasets. On average, SCFDM_{all} is at least 11.58× faster than three baselines for each dataset. Note that for datasets where baselines exceeded the time limit, we conservatively recorded the time as 32, 400s (TL); the actual execution times are likely significantly higher. Specifically, it achieved a maximum speed-up of 130.78× on the Adult dataset compared to CTANE. On Syn-1, where all three baselines triggered the time limit, SCFDM_{all} successfully completed the task in 4, 682.95s, demonstrating its scalability in handling large search spaces. Figure 3(b) shows that SCFDM reduces memory consumption. Baselines suffer from severe memory pressure due to the expansion of lattices in the large search space, averaging peak memory footprints of 149.27GB (CTANE), 132.48GB (Itemset-First), and 135.33GB (FD-First) across all datasets, with CTANE peaking at 195.41GB on Crop. In contrast, SCFDM_{all} cuts the average peak memory footprint to merely 12.96GB, delivering an 10.73× memory reduction on average. This validates that AttrFinder and RepSampler effectively reduce the memory footprint of constant and variable CFD discovery.

The reported runtime of SCFDM includes both data pre-processing and rule discovery, which includes the neural embedding

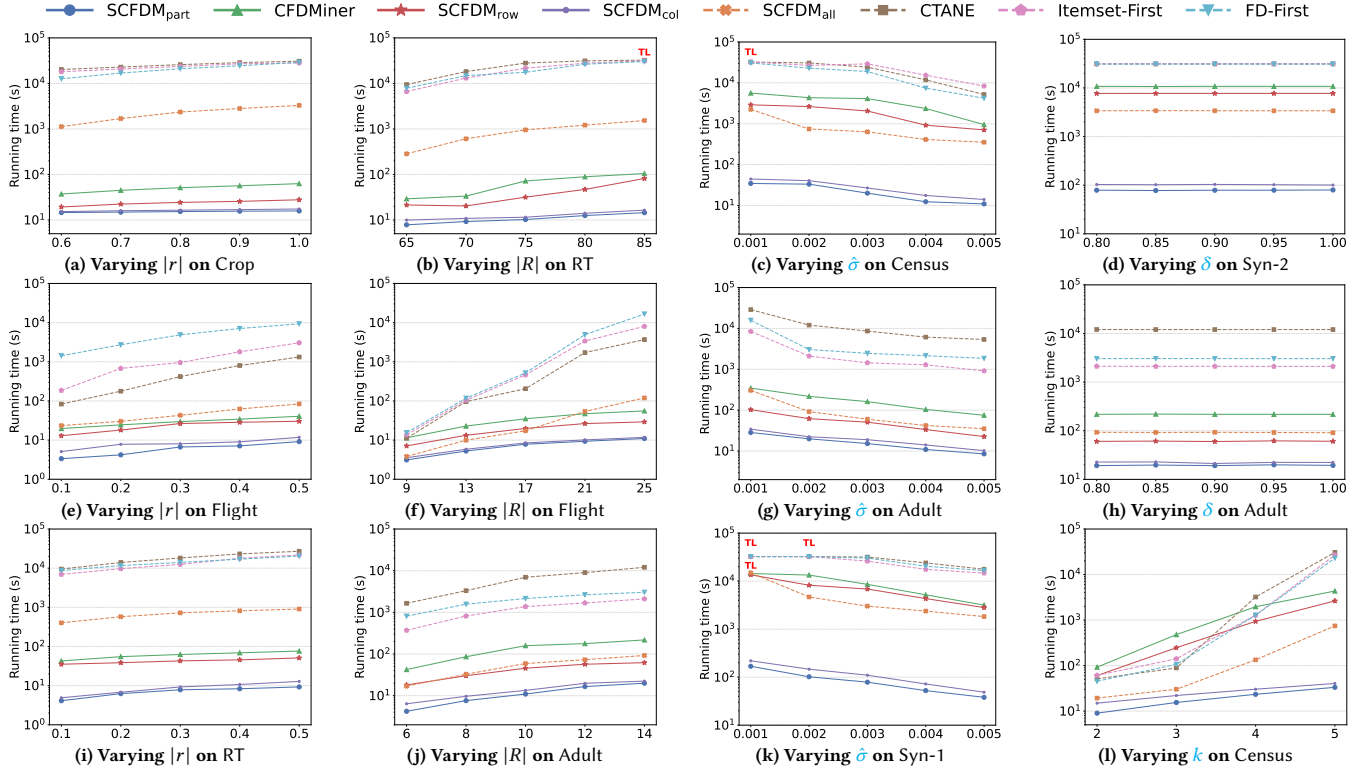


Figure 4: Scalability evaluation of SCFDM

training, correlation extraction, and data partitioning. Empirically, the overhead on embedding and Transformer training is trivial, requiring only 7.42–23.15s (13.17s on average). This overhead accounts for 29.90% of runtime in constant CFD discovery and just 0.55% in both constant and variable CFD discovery on average, dropping to 0.27% on large-scale datasets such as Crop. This shows that the embedding and Transformer training overhead becomes increasingly negligible as the search space grows. Detailed experimental results are provided in Appendix F of [1].

Analysis. There are two main reasons for the high efficiency of SCFDM: (1) partitioning relation r into sub-tables for parallel CFD discovery, which prunes the search space and reduces runtime and memory consumption; and (2) leveraging AttrFinder and RepSampler to identify correlated attributes and representative tuples, thereby reducing redundant attribute and tuple traversal. Moreover, the embedding and Transformer training overhead is modest. In contrast, baselines (e.g., CTANE) lack Transformer-based correlation extraction, resulting in a larger search space.

Exp-2: Scalability of SCFDM. We evaluated SCFDM’s scalability by varying (1) the tuple sampling ratio $|r|$, (2) the number of attributes $|R|$, (3) the normalized support threshold $\hat{\sigma}$, (4) the confidence threshold δ , (5) the maximum CFD size k , which limits the number of involved attributes (LHS and RHS combined) in a CFD, and (6) the number of CPU cores q .

Varying $|r|$. As shown in Figures 4(a), 4(e) and 4(i), the tuple sampling ratio $|r|$ was tuned from 0.6 to 1.0 on Crop and from 0.1 to 0.5 on Flight and RT. As $|r|$ grew, the running times of SCFDM_{part}, SCFDM_{all}, SCFDM_{row}, SCFDM_{col} along with the baselines increased due to the expanded search space. Across the three datasets, SCFDM_{part} achieved an average efficiency gain of 4.96×

over CFDMiner. Meanwhile, SCFDM_{all} outperformed the baselines (CTANE, Itemset-First, and FD-First) with an average speedup of 13.77×. Specifically, SCFDM_{part} achieved a 3.32× efficiency gain over CFDMiner on Crop, while SCFDM_{all} exhibited higher improvements, outperforming the baselines by 47.88× and 22.62× on Flight and RT on average, respectively. We vary the sampling ratio on Crop, Flight, and RT, which are sufficiently large for changes in the sampling ratio to substantially affect the sampled workload. The other datasets exhibit consistent trends under different sampling ratios (see Appendix H.6 of [1]).

Varying $|R|$. Figures 4(b), 4(f), 4(j) show the impact of varying attribute number $|R|$ across three datasets: RT (65–85), Flight (9–25), and Adult (6–14). As $|R|$ increased, the runtimes of all baselines escalated significantly. In contrast, SCFDM_{part}, SCFDM_{row}, SCFDM_{col}, and SCFDM_{all} exhibited much more resilient and gradual growth. Specifically, SCFDM_{part} outperformed CFDMiner by 11.51× on Adult, while SCFDM_{all} achieved higher speedups, which beat the baselines by an average of 23.00×, 65.03× and 60.24× on RT, Flight and Adult, respectively. Figure 4(b) highlights that SCFDM_{col} outperforms SCFDM_{row} in speedup. These datasets cover wide (RT, 65–85), medium (Flight, 9–25), and narrow (Adult, 6–14) schemas, providing coverage across different numbers of attributes $|R|$. The remaining datasets exhibit similar trends as $|R|$ varies (see Appendix H.6 of [1]).

Varying $\hat{\sigma}$. Figures 4(c), 4(g), 4(k) illustrate the impact of varying the normalized support threshold $\hat{\sigma}$ (as defined in Section 7.1) from 1×10^{-3} to 5×10^{-3} across Census, Adult and Syn-1. As $\hat{\sigma}$ increased, all baselines saw a significant decrease in execution time due to more candidate pruning. Specifically, SCFDM_{part} and SCFDM_{all} consistently outperformed the baselines, achieving average speedups of 156.58× (resp. 22.84×) on Census, 10.91× (resp.

Table 3: Parallel performance of SCFDM with varying $|q|$

Algorithm	Dataset	Execution Time (s)			
		$q = 28$	$q = 56$	$q = 84$	$q = 112$
SCFDM _{part}	RT	35.41	22.23	17.80	14.48
SCFDM _{all}	RT	4,904.92	3,422.11	2,654.12	2,031.53
SCFDM _{part}	Census	75.33	52.13	41.03	33.27
SCFDM _{all}	Census	1,689.25	1,274.70	908.41	744.68
SCFDM _{part}	Adult	45.35	32.61	26.62	19.93
SCFDM _{all}	Adult	217.08	133.11	102.66	92.18
SCFDM _{part}	Syn-1	188.32	159.78	117.84	101.85
SCFDM _{all}	Syn-1	12,240.28	8,201.08	6,102.75	4,682.95

62.82 $\times$) on Adult, and 101.70 $\times$ (resp. 4.90 $\times$) on Syn-1.

Varying δ . In Figures 4(d) and 4(h), we found that the execution times of SCFDM, CTANE, Itemset-First, and FD-First showed little change when varying the confidence threshold δ from 0.80 to 1.00. This is because δ has a limited effect on the search space. Specifically, SCFDM_{part} and SCFDM_{all} achieved average speedups of 136.49 $\times$ (resp. 9.19 $\times$) on Syn-2 and 11.20 $\times$ (resp. 62.16 $\times$) on Adult. *Since changing δ has little effect on the candidate space, the performance of all algorithms changes only slightly. We therefore show two representative plots, while the remaining datasets exhibit the same trend (Appendix H.6 of [1]).*

Varying k . We varied the maximum CFD size k from 2 to 5 on Census. Figure 4(l) shows that as k increased, the running times of SCFDM, CTANE, Itemset-First, and FD-First grew exponentially, with SCFDM growing more slowly. On average, SCFDM_{part} and SCFDM_{all} were 84.77 $\times$ and 31.14 $\times$ more efficient, respectively. See Appendix H.3 in [1] for the setting guidance of k . *Census is chosen as it has the largest number of attributes among the datasets on which all baselines complete, making the effect of increasing k on the search space more pronounced.*

Varying q . We increased the number of CPU cores q from 28 to 112 to assess the scalability of both SCFDM_{part} and SCFDM_{all} on the RT, Census, Adult, and Syn-1 datasets. Table 3 shows that SCFDM scales effectively with more CPU cores. As q increased, the average runtime of SCFDM across the four datasets decreased from 2,424.5s to 965.1s. Specifically, SCFDM_{part} and SCFDM_{all} achieved an average speedup of 2.03 $\times$ and 2.52 $\times$ across the four datasets, respectively. *We provide the full results in Appendix H.6 of [1], and the remaining datasets show the same trends.*

Analysis. The experimental results demonstrate the effectiveness of SCFDM. (1) Its execution time varies most with $|R|$ and $|k|$, due to their exponential effect on the search space of sub-tables. (2) SCFDM scales well with parameters $|r|$ and $|\hat{\sigma}|$ by sampling a fixed number of representative tuples and efficiently discovering CFDs on sub-tables. (3) The parameter $|\delta|$ has little impact on the search space. (4) Additionally, increasing the number of CPU cores $|q|$ speeds up SCFDM by distributing the workload across CPU cores with low communication overhead.

Exp-3: Effectiveness of SCFDM. We evaluated the effectiveness of SCFDM on all seven datasets. As shown in Table 4, SCFDM consistently achieves the highest F_1 -score across all datasets. While all methods generally maintain high precision, they exhibit different performances on recall. Specifically, CFDMiner shows the lowest recall (e.g., only 0.062 on Census). Although it operates with relatively high efficiency, it is limited to discovering constant CFDs, thus missing a large number of variable CFDs. As for CTANE, Itemset-First, and FD-First, the low F_1 -scores stem from (1) a large search space,

Table 4: Effectiveness of CFD discovery

Dataset	Metric	CFDMiner	CTANE	Itemset-First	FD-First	SCFDM
RT	Precision	0.985	0.984*	0.974	0.972	0.986
	Recall	0.075	0.709*	0.966	0.962	0.960
	F_1	0.139	0.824*	0.970	0.967	0.973
Census	Precision	0.982	0.952	0.958	0.966	0.987
	Recall	0.062	0.990	0.988	0.991	0.983
	F_1	0.117	0.971	0.973	0.978	0.985
Crop	Precision	0.970	0.952	0.946	0.950	0.971
	Recall	0.084	0.949	0.952	0.949	0.933
	F_1	0.155	0.950	0.949	0.949	0.952
Flight	Precision	0.999	0.996	0.995	0.995	0.997
	Recall	0.140	0.998	0.999	0.999	0.998
	F_1	0.246	0.997	0.997	0.997	0.997
Adult	Precision	1.000	1.000	1.000	1.000	1.000
	Recall	0.338	1.000	1.000	1.000	1.000
	F_1	0.505	1.000	1.000	1.000	1.000
Syn-1	Precision	0.977	0.979*	0.960*	0.966*	0.979
	Recall	0.041	0.833*	0.872*	0.868*	0.953
	F_1	0.079	0.900*	0.914*	0.914*	0.966
Syn-2	Precision	0.985	0.979	0.964	0.960	0.980
	Recall	0.079	0.962	0.970	0.970	0.966
	F_1	0.146	0.970	0.967	0.965	0.973

Results obtained before reaching the time (9 hours)/memory (224GB) limit.

* indicates that the algorithm timed out before completion. **Bold** indicates the best performance.

which triggers the 9-hour timeout limit on large-scale datasets (e.g., Syn-1). and (2) the absence of attribute-correlation analysis, which introduces spurious CFDs and consequently degrades precision. In contrast, SCFDM strikes a balance by reducing the search space through attribute-aware pruning and representative sampling, achieving high accuracy while maintaining strong scalability. Specifically, on the RT and Syn-1 datasets, SCFDM achieves highest F_1 -scores of 0.973 and 0.966, respectively, surpassing all baselines. To ensure a fair evaluation for CFDMiner, we provide a comparison restricted to constant CFD discovery in Appendix H.1 in [1].

Exp-4: Sensitivity to CFD Parameters. We evaluated the impact of CFD parameters (support thresholds $\hat{\sigma}$) on the RT and Syn-1 datasets based on the F_1 -score. As shown in Table 5, SCFDM demonstrates robustness to variations in support thresholds. Specifically, when the threshold is small (e.g., $\hat{\sigma} = 1 \times 10^{-3}$), the F_1 -scores of all baselines drop sharply, with certain methods falling below 0.5 (e.g., 0.082 for CFDMiner and 0.488 for CTANE on RT). This is because lower support thresholds expand the search space, causing baselines to trigger timeouts and fail to discover the full set of CFDs. In contrast, SCFDM consistently maintains high F_1 -scores (above 0.90) even at $\hat{\sigma} = 1 \times 10^{-3}$. As $\hat{\sigma}$ increases to 5×10^{-3} , the performance gap narrows since the search space becomes more manageable for baselines. However, SCFDM still yields the best or competitive results across all cases. These results demonstrate the robustness of SCFDM under low support thresholds, consistently achieving high discovery accuracy. Due to space limitations, we present the parameter sensitivity results on the other five datasets in Appendix H.2 of [1]. *The other five datasets show similar trends, as reported in Appendix H.2 of [1].*

Exp-5: Ablation Study of SCFDM. To ensure fair runtime comparison, we conduct ablation studies on a random sample of 80,000 rows from Crop, ensuring that all variants complete execution within the time limit. As shown in Table 6, all three components are essential for balancing quality and efficiency. Omitted RepSampler elevates the runtime to 3,680.8s. *Notably, omitting AttrFinder or RepSampler slightly increases recall (to 99.0% and 98.4%, respec-*

Table 5: Impact of normalized support thresholds $\hat{\sigma}$ on F_1 -score across different algorithms

Dataset	Algorithm	Metric	1×10^{-3}	2×10^{-3}	3×10^{-3}	4×10^{-3}	5×10^{-3}
RT	CFDMiner		0.082	0.115	0.148	0.172	0.184
	CTANE		0.488*	0.824*	0.965	0.970	0.972
	Itemset-First F_1		0.523*	0.970	0.962	0.969	0.976
	FD-First		0.534*	0.967	0.962	0.972	0.982
	SCFDM		0.901	0.970	0.974	0.978	0.982
Syn-1	CFDMiner		0.076	0.079	0.095	0.114	0.119
	CTANE		0.372*	0.900*	0.926*	0.962*	0.977
	Itemset-First F_1		0.454*	0.914*	0.963	0.971	0.972
	FD-First		0.423*	0.914*	0.965	0.968	0.970
	SCFDM		0.927	0.952	0.966	0.971	0.977

Results obtained before reaching the time (9 hours)/memory (224GB) limit.
* indicates that the algorithm timed out before completion.

tively) due to exhaustive exploration over the complete relation, but lowers precision from 98.6% to 96.5% and 97.8%, respectively, by introducing spurious attribute dependencies and redundant CFDs. Algorithm 3 (Line 8) verifies support and confidence on r (support $\geq \sigma$, confidence $\geq \delta$). However, satisfying these thresholds alone does not guarantee that a CFD belongs to the ground-truth set Σ , which is further validated by experts (Section 9.1). Furthermore, PCFDFinder is critical for efficiency. The w/o PCFDFinder variant processes sub-tables sequentially on a single thread while keeping all other components unchanged. Its accuracy therefore remains the same as the full model, but the loss of parallel processing increases the runtime to 18,621.3s.

Evaluating alternative components highlights the effectiveness of SCFDM. First, replacing AttrFinder with Pearson correlation coefficient [35] (computing pairwise attribute correlation between all attribute pairs), XGBoost [15] or Kamino [39] compromises effectiveness as they fail to extract determinant antecedent sets for target attributes. For instance, XGBoost produces redundant attribute combinations, reducing its F_1 -score to 79.8% (with recall dropping to 68.2%) and causing a runtime of 11,025.1s. Second, substituting RepSampler with random [64] or stratified [62] sampling yields only a limited reduction in sampling time (down to 642.2s) but at a heavy degradation in discovery quality, where the F_1 -scores of the two methods drop from 98.2% to 86.4% and 88.2%, respectively, due to severe recall loss. This underscores that alternative sampling strategies cannot sufficiently preserve the tuples essential for CFD discovery.

Overall, these results demonstrate that the combined effects of AttrFinder, RepSampler, and PCFDFinder is essential for SCFDM to achieve both accuracy and runtime scalability.

Exp-6: Case study on data repair. We conduct a case study on the noisy Adult dataset to evaluate SCFDM’s effectiveness in data repair. For example, some tuples with “Relationship = Husband” are mislabeled as “Female” in “Gender”, while others contain missing values (marked as “?”). SCFDM identifies rules such as φ_5 : ([Education, Occupation] $\rightarrow$ Workclass, (some-college, _ | _)) and φ_6 : ([Gender, Hours, Occupation] $\rightarrow$ Marital, (male, 50, sales | married-civ-spouse)). For example, φ_5 captures the association between “some-college” education and workclass, enabling us to identify and repair inconsistent workclass values. Meanwhile, φ_6 indicates that men aged 50 or above who work 50 or more hours per week in sales are likely to have Marital = married-civ-spouse. In Adult, Age and Hours values above 50 are recorded as 50. This pattern is also reasonable, as long working hours among senior individuals can be associated with family and household responsibilities [3]. We use φ_6 to repair tuples violating this pattern.

Table 6: Ablation study of key SCFDM components on Crop

Model Configuration	Prec. (%)	Rec. (%)	F1 (%)	Runtime (s)
SCFDM (Full Model)	98.6	97.9	98.2	655.4
Ablations				
w/o AttrFinder	96.5	99.0	97.7	28,435.7
w/o RepSampler	97.8	98.4	98.1	3,680.8
w/o PCFDFinder	98.6	97.9	98.2	18,621.3
Alternative Components				
w/o AttrFinder + w/ Pearson [35]	97.0	57.4	72.1	5,923.1
w/o AttrFinder + w/ XGBoost [15]	96.1	68.2	79.8	11,025.1
w/o AttrFinder + w/ Kamino [39]	98.0	76.5	85.9	4,388.2
w/o RepSampler + w/ random [64]	97.5	77.6	86.4	642.2
w/o RepSampler + w/ stratified [62]	98.2	80.1	88.2	644.6

A quantitative evaluation (Appendix H.5 of [1]) shows that the discovered CFDs correct 912 of 1,000 injected errors, with 94.6% precision, 91.2% recall, and $F_1 = 0.929$. In contrast, baselines suffer from semantic noise and lattice explosion, producing spurious and redundant CFDs while missing correlated rules such as φ_5 and φ_6 . Without attribute correlation analysis, irrelevant candidates overwhelm the search space, hindering CFD discovery for data cleaning on large datasets.

Summary. We find: (1) Efficiency. SCFDM outperforms all baselines. It achieves an average runtime speedup of 105.18 $\times$ in constant CFD discovery (up to 136.49 $\times$ on Syn-2) and at least 11.58 $\times$ in both constant and variable CFD discovery. Furthermore, SCFDM reduces average peak memory to 5.11GB (5.13 $\times$ reduction) for constant CFD discovery and 12.96GB (10.73 $\times$ reduction) for both constant and variable CFD discovery, ensuring high memory efficiency. (2) Robust scalability. SCFDM scales robustly across all parameters, maintaining gradual runtime growth where baselines grow exponentially with increasing $|R|$ or $|k|$. (3) High discovery quality. SCFDM consistently achieves the highest F_1 -scores (above 0.90), balancing precision and recall better than all baselines. (4) Component vitality. Ablation confirms that AttrFinder, RepSampler, and PCFDFinder are indispensable, as they together outperform alternatives (Pearson correlation, XGBoost, Kamino, Random sampling, and Stratified sampling) by filtering semantic noise and spurious candidate CFDs. (5) Efficient parallelism. SCFDM scales efficiently with computational resources, reducing average runtime from 2,424.5s to 965.1s and achieving a 2.03 $\times$ to 2.52 $\times$ speedup when the number of CPU cores $|q|$ increases from 28 to 112. (6) Practical utility. The case study and quantitative repair evaluation (Appendix H.5 of [1]) show that SCFDM discovers high-quality CFDs for effective data repair, while baselines produce spurious CFDs and miss valid constraints such as φ_5 and φ_6 within the time limits.

10 CONCLUSION

We propose SCFDM, a data-parallel framework for efficient CFD discovery on large-scale datasets. Its key innovations include: (1) Probabilistic representation: Modeling CFDs as conditional probabilities and learning attribute correlations with a two-stage Transformer. (2) Attention-driven extraction: AttrFinder derives correlated attribute sets from global attention to partition the relation into sub-tables. (3) Diversity-aware sampling: RepSampler reduces the input size while preserving discovery recall through a theoretically grounded sampling bound. (4) Data parallelism: Independent sub-tables are mined in parallel for scalable CFD discovery.

Our future work will focus on extending SCFDM to support more complex rules with machine learning predicates (e.g., REEs [33]).

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A TAXONOMY OF CFD PATTERN TUPLES

Conditional functional dependencies (CFDs) extend functional dependencies (FDs) by embedding pattern tuples to enforce data constraints. A pattern tuple t_p is defined over the attribute set $X \cup \{Y\}$ for a dependency $X \rightarrow Y$. We categorize CFDs into constant and variable CFDs. This decomposition is supported by Lemma 1 of [30], which shows that any CFD set Σ can be equivalently partitioned into a constant subset Σ_c and a variable subset Σ_v ($\Sigma \equiv \Sigma_c \cup \Sigma_v$).

Constant CFD patterns. A constant CFD pattern tuple assigns a constant value to every attribute, i.e., $t_p[B] = c$ for each $B \in X \cup \{Y\}$, where $c \in \text{dom}(B)$. When the consequent Y is fixed to a constant value, any antecedent attribute in X assigned to an unnamed variable “_” becomes unnecessary and can be removed without affecting minimality [30]. Consequently, any CFD with a constant RHS can be simplified into a constant form containing only constant attributes in $X \cup \{Y\}$. Such CFDs represent instance-level rules and are commonly used in data analysis and data cleaning.

Variable CFD patterns. A CFD is a variable CFD if its pattern tuple contains one or more unnamed variables “_”. According to the locations of constants and variables, variable CFDs can be categorized into two types. The first type is the pure variable CFD, where $t_p[B] = _$ for every attribute $B \in X \cup \{Y\}$. In this case, the CFD reduces to a functional dependency (FD) that holds for all tuples in the relation r . The second type is the mixed variable CFD. In this pattern, the antecedent X contains both constants and unnamed variables (i.e., $X^c \neq \emptyset \wedge X^v \neq \emptyset$), while the consequent remains an unnamed variable ($t_p[Y] = _$). Unlike pure variable CFDs, mixed variable CFDs describe dependencies that hold only under specific conditions defined by the antecedent constants. For example, the pattern tuple (UK, _ || _) over the dependency (Country, ZipCode $\rightarrow$ City) indicates that the CFD is considered only for tuples where Country = UK.

We note that, for a minimal mixed variable CFD, the consequent $t_p[Y]$ cannot be a constant. If $t_p[Y] = c$, then, according to the semantics of CFDs [30], any antecedent attribute assigned to an unnamed variable “_” no longer contributes additional constraints to the matching tuples. Such attributes therefore become redundant and can be removed without affecting the dependency. As a result, the mixed pattern is simplified into a constant CFD pattern. Hence, to preserve the mixed-variable structure and minimality, the consequent Y must remain an unnamed variable “_”.

B EXTENSION AND ADAPTABILITY FOR PATTERN TABLEAU-BASED CFDs

Following Definition 3.3, this section extends SCFDM to support the discovery of tableau-based CFDs ($\varphi_{\mathcal{T}}$). SCFDM can be extended to support pattern tableau with slight modifications, with only changes in PCFDFinder and no changes to the rest of the system.

Impact on AttrFinder. The main concern is whether the attribute correlation extraction step might miss constant relationships (e.g., $X = A \vee X = B \rightarrow Y = C$). We clarify that this does not affect AttrFinder. In our Transformer-based design, the self-attention

mechanism learns the conditional relationship $P_r(Y|X)$ through masked token reconstruction. If a tableau-based CFD holds in the data, it means that multiple different constant values of X can strongly predict Y . As a result, during training over the full tuple space, the attention weights will become large for the corresponding (X, Y) attribute pairs. Therefore, AttrFinder can still correctly capture such correlated attributes and group them into the same correlated set, without requiring any changes to the algorithm.

Impact on RepSampler. Another key concern is whether representative sampling might miss different constant branches needed to build a tableau $\mathcal{T}_p$. We show that RepSampler addresses this issue through its diversity-first design. Specifically, locality-sensitive hashing (LSH) groups tuples into different buckets based on their similarity in the embedding space. Tuples satisfying $X = A$ and those satisfying $X = B$ tend to fall into different buckets. By sampling tuples across all non-empty buckets, RepSampler ensures that all groups are fairly represented, including those corresponding to different constant patterns. As a result, the sampled data preserves the tuple diversity needed for tableau construction, without requiring any changes to the algorithm.

Adaptation for PCFDFinder. PCFDFinder can be extended to support tableau-based CFDs in two simple ways. A straightforward approach is to keep existing single-pattern CFD miners (e.g., CFDMiner) and add a post-processing step after parallel mining. In this step, CFDs that share the same attribute set $X \rightarrow Y$ are merged into a single pattern tableau $\mathcal{T}_p$. This only introduces linear overhead, i.e., $O(|\Sigma|)$, where $|\Sigma|$ is the number of discovered CFDs. Alternatively, the base serial miner can be directly replaced with any tableau-based discovery method, such as those in [41].

Incorporating pattern tableau into existing discovery algorithms leads to an exponential increase in the search space, making the problem intractable for large-scale relations. This challenge further highlights the advantage of SCFDM. By applying AttrFinder, SCFDM partitions the original relation into smaller sub-tables of highly correlated attributes, significantly reducing the search space before mining. Together with the linear-time $O(|\Sigma|)$ post-processing step, SCFDM makes the discovery of tableau-based CFDs both efficient and scalable on large datasets.

C PROOF OF THEOREM 1 AND THEOREM 2

We prove Theorem 1 and Theorem 2 as follows.

C.1 Proof of Theorem 1

The ratio r_c of the variance-augmented bound to Chebyshev’s bound can be derived as:

$$r_c = \frac{\frac{\delta(1-\delta) \ln(\frac{2}{1-\delta\beta})}{2\epsilon^2}}{\frac{\delta(1-\delta)}{\epsilon^2(1-\delta\beta)}} = \frac{(1-\delta\beta) \ln(\frac{2}{1-\delta\beta})}{2}.$$

Since $0 < \delta\beta \leq 1$, the function $f(u) = u \ln(2/u)$ (where $u = 1 - \delta\beta$) reaches its maximum near $u = 0.73$ with $f(u) \approx 0.74 < 2$. Thus, $0 < r_c < 1$, proving it is tighter than Chebyshev’s bound.

Similarly, to analyze the ratio r_b relative to Bennett’s bound, we leverage the inequality $\ln(1+x) > \frac{2x}{2+x}$ and the Taylor expansion

$(\hat{\sigma} + \epsilon) \ln(1 + \frac{\epsilon}{\hat{\sigma}}) - \epsilon \approx \frac{\epsilon^2}{2\hat{\sigma}}$ for small ϵ , yielding:

$$r_b = \frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{2}{1-\delta\beta})}{2\epsilon^2} \cdot \frac{(\hat{\sigma} + \epsilon) \ln(1 + \frac{\epsilon}{\hat{\sigma}}) - \epsilon}{\ln(\frac{2}{1-\delta\beta})} \\ \approx \frac{\hat{\sigma}(1-\hat{\sigma})}{2\epsilon^2} \cdot \frac{\epsilon^2}{2\hat{\sigma}} = \frac{1-\hat{\sigma}}{4} < 1.$$

This concludes that $0 < r_b < 1$, confirming it is tighter than Bennett's bound. $\square$

C.2 Proof of Theorem 2

The proof consists of the following three parts.

(1) *Size of samples.* The ratio r of the variance-augmented bound to the hybrid Chernoff bound [17] is expressed as:

$$r = \frac{\frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{1-\delta\beta}{2})}{-2\epsilon^2}}{\frac{3\tau}{\epsilon^2} \ln(\frac{2}{1-\delta\beta})} = \frac{\hat{\sigma}(1-\hat{\sigma})}{6\tau}.$$

The regime classification boundaries $\hat{\sigma}_1$ and $\hat{\sigma}_2$ are determined by solving the threshold equation $\frac{\hat{\sigma}(1-\hat{\sigma})}{6\tau} = 1$, which yields:

$$\hat{\sigma}_1 = \frac{1 - \sqrt{1 - 24\tau}}{2}, \quad \hat{\sigma}_2 = \frac{1 + \sqrt{1 - 24\tau}}{2},$$

for $\tau \leq \frac{1}{24}$. The condition $\frac{\hat{\sigma}(1-\hat{\sigma})}{6\tau} < 1$ ensures that the variance-augmented Hoeffding-Bernstein inequality provides a tighter lower bound. Thus, the bound $N = \frac{\hat{\sigma}(1-\hat{\sigma}) \ln(\frac{2}{1-\delta\beta})}{2\epsilon^2}$ holds for the following intervals respectively:

$$\begin{cases} 0 \leq \hat{\sigma} \leq \hat{\sigma}_1 \text{ and } \hat{\sigma}_2 < \hat{\sigma} \leq 1, & \text{when } \tau \leq \frac{1}{24} \\ 0 \leq \hat{\sigma} \leq 1, & \text{when } \tau > \frac{1}{24}. \end{cases}$$

Otherwise, the hybrid Chernoff bound $N = \frac{3\tau}{\epsilon^2} \ln(\frac{2}{1-\delta\beta})$ is selected.

(2) *Support threshold.* Under random sampling, the occurrence count of any pattern in the sample follows a standard sampling distribution (e.g., hypergeometric or binomial) determined by the original relation r [77]. As a result, the normalized support $\hat{\sigma}'$ measured in the sample relation r_s provides an unbiased estimate of the true support $\hat{\sigma}$ in the full relation r . In other words, random sampling preserves the expected support of a pattern, which can be formally expressed as:

$$\mathbb{E}[\hat{\sigma}'] = \hat{\sigma}.$$

This result implies that, under the additive estimation error ϵ established in Part (1), the sample support $\hat{\sigma}'$ is highly likely to remain close to the true support $\hat{\sigma}$. Specifically, the probability that their difference exceeds ϵ is bounded by $1 - \delta$. Therefore, using $\hat{\sigma}'$ as an estimate of $\hat{\sigma}$ introduces no systematic bias and preserves the validity of the subsequent rule discovery process.

(3) *Confidence threshold.* To ensure that a constraint discovered in the sample relation r_s with confidence δ' still satisfies the target confidence δ in the original relation r , we compensate for the sampling error. Let P denote the true confidence of $\mathbf{X} \rightarrow Y$ in r , and let P_s denote the confidence observed in r_s . According to the error bound derived above, the difference between P_s and P is bounded by ϵ with probability at least δ , i.e., $\mathbb{P}(|P_s - P| \leq \epsilon) \geq \delta$.

To account for the worst-case effect of sampling noise, we analyze how the confidence estimated from the sample may deviate

from its true value in the original relation. Under the Bernoulli sampling model [75], the variance of the empirical confidence estimator is given by $\sigma_{sample}^2 = \frac{P(1-P)}{N}$, where P is the true confidence and N is the sample size. Let $\Delta P = P_s - P$ denote the estimation error introduced by sampling. Since Part (1) guarantees that $|\Delta P| \leq \epsilon$, we further examine how this bounded error affects the confidence threshold. For small values of ϵ , a first-order Taylor approximation [38] provides an accurate estimate of this effect. Specifically, we expand the confidence evaluation function around the threshold δ as:

$$f(P_s) \approx f(P) + \frac{\partial f}{\partial P} \Delta P + O(\Delta P^2).$$

To account for the worst-case impact of sampling noise, we derive a mapping from the target confidence threshold δ to the adjusted sample confidence threshold δ' , given by:

$$\delta = \delta' \cdot \left(1 + \frac{\epsilon}{2} Var(\mathbf{X})\right) \approx \delta' \cdot \frac{1 + \frac{\epsilon}{4}}{1 - \frac{\epsilon}{4}},$$

where the maximum variance component $Var(\mathbf{X})$ under random assignment non-trivially approaches the tightly bounded constant scaling of $\frac{1}{2}$. Rearranging the algebraic terms to isolate the conservative sample threshold δ' , we obtain:

$$\delta' = \frac{1 - \frac{\epsilon}{4}}{1 + \frac{\epsilon}{4}} \delta.$$

This calibration depends only on the statistical properties of sampling error and is independent of the specific CFD discovery strategy (e.g., BFS-based lattice traversal). As a result, it can be applied universally across different CFD discovery algorithms. $\square$

D TRAINING DETAILS IN AttrFinder

The training of AttrFinder is a self-supervised process designed to extract dependencies through masked reconstruction over the tensor $\mathcal{T}$. Given a batch of B embedded tuples $\mathcal{T}_B \in \mathbb{R}^{B \times m \times d}$, where $B \leq n$, the training procedure follows four critical phases.

Stochastic attribute masking. To compel the model to learn conditional dependencies $P(Y|X)$, we apply a masking operator $\tilde{\mathcal{T}}_B = \text{Mask}(\mathcal{T}_B)$. For each tuple t_j in the batch, we randomly select a subset of attributes and replace their d -dimensional embeddings $E_i^{(j)}$ with a learnable `[MASK]` token. This simulates the discovery of unknown consequents at scale, forcing the model to reconstruct the masked attributes by attending to the remaining contextual attributes across diverse tuple instances.

Multi-Head attention interaction (Encoder). The masked batch $\tilde{\mathcal{T}}_B$ is processed through L layers. In each layer $l \in [1, L]$, the self-attention mechanism computes interaction matrices for each tuple t_j :

$$A_{l,h}^{(j)} = \text{softmax}\left(\frac{Q_{l,h}^{(j)} (K_{l,h}^{(j)})^T}{\sqrt{d_k}}\right),$$

where Q, K are linear projections of the j -th tuple's representation from the previous layer, and $A_{l,h}^{(j)} \in \mathbb{R}^{m \times m}$ captures the tuple-specific correlations.

Segmented categorical reconstruction. The final hidden state of the L -th layer, $\mathbf{H}^{(L)} \in \mathbb{R}^{m \times d}$, is passed through a linear projection layer. For each masked attribute A_i with c_i categories, we

apply a segmented softmax to reconstruct the original zero-one distribution:

$$\hat{y}_{i,k} = \frac{\exp(W_{out}^{(i)} \mathbf{h}_i^{(L)} + b_{out}^{(i)})_k}{\sum_{p=1}^{c_i} \exp(W_{out}^{(i)} \mathbf{h}_i^{(L)} + b_{out}^{(i)})_p}.$$

This ensures the output for each attribute segment represents a probability distribution, where $\hat{y}_{i,k}$ approximates the conditional probability $P(A_i = v_k \mid attr(\mathcal{R}) \setminus \{A_i\})$.

Optimization and convergence. The model parameters $\Theta = \{\Theta_{emb}, \Theta_{attn}, \Theta_{out}\}$ are optimized end-to-end. Specifically, the embedding parameters $\Theta_{emb} = \{W_{emb}^{(i)}, \mathbf{b}_{emb}^{(i)}\}_{i=1}^m$ are updated simultaneously with the attention and projection weights. We minimize the batch-averaged cross entropy loss $\mathcal{L}$ aggregated over all masked positions across B tuples:

$$\mathcal{L} = -\frac{1}{B} \sum_{j=1}^B \sum_{i \in \text{Masked}_j} \sum_{k=1}^{c_i} y_{i,k}^{(j)} \log(\hat{y}_{i,k}^{(j)}),$$

where $y_{i,k}^{(j)}$ is the ground-truth label for the i -th attribute of the j -th tuple. This global objective ensures that the gradients $\nabla_{\Theta} \mathcal{L}$ reflect the statistical consensus of the entire relation r . During each training step, the gradients are backpropagated from the reconstruction layer through the Transformer layers to the attribute embedding layer. The update rule for the embedding parameters at iteration t using the Adam optimizer is:

$$\Theta_{emb}^{(t+1)} = \Theta_{emb}^{(t)} - \eta \cdot \text{Adam}(\nabla_{\Theta_{emb}} \mathcal{L}),$$

where η is the learning rate (e.g., 5×10^{-4}). This optimization ensures that the dense representations E_i in the latent space are tuned to minimize uncertainty in attribute reconstruction. Through these iterative updates, the attention maps $\mathbf{A}$ and the embeddings Θ_{emb} are updated, eventually converging to a state that reflects the underlying conditional functional dependencies of the relation r .

E COMPLEXITY OF AttrFinder

To evaluate the scalability of AttrFinder, we analyze its computational complexity across three main stages and compare it with traditional discovery methods. This comparison highlights the efficiency advantages of our neural approach.

Data Encoding. The first stage converts the relation r into a tensor representation $\mathcal{T} \in \mathbb{R}^{n \times m \times d}$. Scanning the n tuples to identify distinct values and perform one-hot encoding requires $O(n \cdot m)$. The resulting sparse vectors are then projected into a d -dimensional embedding space, which also takes $O(n \cdot m)$ when treating d as a constant. Therefore, the overall complexity of this stage is linear in the size of the relation, making it efficient for large-scale datasets.

Transformer-based learning. This is the most computationally intensive phase, where the model captures attribute correlations across the dataset. For each of the n tuples, the self-attention mechanism computes an $m \times m$ interaction matrix. Across all epochs and layers, the training complexity is $O(n \cdot m^2)$ when treating the number of epochs, layers, and embedding dimension as constants. Although these constants affect absolute runtime, the growth remains quadratic in the number of attributes m , far more efficient than the exponential $O(2^m)$ cost of traditional search-based methods. Moreover, the operations are highly parallelizable on GPUs.

Correlated attribute set extraction. Once the model has converged, CFD extraction becomes largely independent of the dataset size n . The first step computes the saliency matrix $\bar{\mathbf{A}}$ by aggregating attention maps across all n tuples and L Transformer layers, resulting in a complexity of $O(n \cdot m^2)$ when L and the number of attention heads H are treated as constants. The second step scans $\bar{\mathbf{A}}$ to identify candidate antecedent sets X for each of the m possible consequents, requiring only $O(m^2)$ time.

F ANALYSIS OF TRANSFORMER TRAINING OVERHEAD IN AttrFinder

To better understand the runtime of SCFDM, this section reports the Transformer training time and its proportion in the total runtime.

Constant CFD discovery (SCFDM_{part}). For constant CFD discovery tasks, the main cost is learning attribute correlations. The Transformer training time ranges from 6.32s to 20.68s across benchmarks, with an average of 11.43s. On average, it accounts for 29.07% of the total runtime. The highest proportion appears on Flight, where training takes 6.32s of 10.98s total runtime (57.56%), followed by RT (54.14%, 7.84s / 14.48s), Crop (52.03%, 8.22s / 15.80s), and Adult (49.97%, 9.96s / 19.93s). As dataset size or tuple diversity increases, the rule discovery phase takes a larger share of runtime, reducing the relative cost of Transformer training. This is reflected in Census (37.06%, 12.33s / 33.27s), Syn-2 (18.54%, 14.64s / 78.96s), and Syn-1 (20.30%, 20.68s / 101.85s), where Transformer training occupies a smaller proportion of the total runtime.

Both constant and variable CFD discovery (SCFDM_{all}). In both constant and variable discovery tasks, the search space grows exponentially, so the Transformer training overhead is amortized by the much larger search cost, reducing its average runtime share to 0.54% across datasets. It accounts for a maximum proportion of 10.98% on the Adult dataset, where it requires 10.12s out of the 92.18s total execution time, followed by 5.49% on Flight (6.55s out of 119.32s). Crucially, on larger datasets or those with higher tuple diversity, where traditional rule discovery algorithms face the most severe execution bottlenecks, the proportion of Transformer training time decreases significantly. This empirical trend is validated across the remaining datasets, where the training consumes 9.82s out of 744.68s on Census (1.32%), 15.43s out of 3,392.30s on Syn-2 (0.45%), 8.48s out of 2,031.53s on RT (0.42%), 18.15s out of 4,682.95s on Syn-1 (0.39%), and reaches its minimum proportion of 0.28% on the Crop dataset, taking 9.16s out of the 3,293.83s total end-to-end time. This scaling behavior demonstrates the scalability and efficiency of SCFDM in handling large search spaces, showing that the Transformer training delivers performance gains as data size and search space increase.

G COMPLEXITY OF RepSampler

To evaluate the efficiency of RepSampler, we analyze its computational complexity across its main operational stages and show that it scales linearly with the number of tuples n .

Hash-based partitioning. This phase organizes the data into structural bucket without the overhead of quadratic comparisons. RepSampler employs an LSH framework based on the MinHash strategy to map n tuples into similarity buckets. Since the algorithm

Table 7: Effectiveness evaluation on constant CFDs

Dataset	Metric	CFDMiner	SCFDM _{part}
RT	Precision	0.980	0.988
	Recall	0.971	0.971
	F_1	0.975	0.979
Census	Precision	0.942	0.985
	Recall	0.977	0.969
	F_1	0.959	0.977
Crop	Precision	0.970	0.984
	Recall	0.965	0.946
	F_1	0.967	0.965
Flight	Precision	0.994	0.999
	Recall	0.996	0.996
	F_1	0.995	0.997
Adult	Precision	1.000	1.000
	Recall	1.000	1.000
	F_1	1.000	1.000
Syn-1	Precision	0.967	0.981
	Recall	0.970	0.964
	F_1	0.968	0.972
Syn-2	Precision	0.980	0.992
	Recall	0.992	0.990
	F_1	0.986	0.991

only processes the m non-zero entries (original attributes) for each of the hash functions, the complexity of this stage is $O(n)$. Unlike iterative clustering (e.g., K-means [61]) or density-based methods (e.g., DBSCAN [26]), which often scale quadratically or require multiple passes, our hashing-based approach achieves partitioning in a single linear scan, effectively bypassing the $O(n^2)$ bottleneck of pairwise distance calculations.

Bucket management. This stage is designed to preserve rare patterns that still represent valid CFDs, even though they have low support. Sorting the k resulting buckets by their population size takes $O(k \log k)$ to facilitate the diversity-first selection logic. In real-world datasets, the number of buckets k is constrained by the LSH parameters and is significantly smaller than the total number of tuples ($k \ll n$), making this overhead negligible while ensuring the sampling is not biased toward dominant, redundant patterns.

Stratified sampling. The final phase extracts the minimal set of tuples required to satisfy the discovery recall guarantee. RepSampler iteratively extracts representative tuples from the managed buckets until the theoretical sampling lower bound N is reached, resulting in $O(N)$. While traditional discovery algorithms must process the full dataset n , RepSampler reduces the input size to a compact representative set N . Since $N \ll n$, RepSampler lowers the computational cost of downstream mining while maintaining a recall guarantee.

H ADDITIONAL EXPERIMENTAL DETAILS

H.1 Evaluation on Constant CFDs in Exp-3.

To avoid evaluation bias caused by CFDMiner’s restriction to constant CFDs, we conduct an additional evaluation focusing on constant CFD discovery. Under this setting, the ground truth contains only constant CFDs, which allows a direct comparison between CFDMiner and SCFDM_{part}.

As reported in Table 7, even when restricted only to the constant CFDs, SCFDM_{part} consistently achieves superior F_1 -scores across most evaluated datasets, with an average F_1 -score of 0.980 and a

peak value of 1.000. In contrast, CFDMiner achieves an average F_1 -score of 0.976 across these datasets. While CFDMiner achieves higher recall than in both constant and variable CFD discovery setting (Table 4), it still exhibits low precision and F_1 scores on large-scale datasets (e.g., Census). Specifically, on the large Census dataset with diverse tuples, CFDMiner achieves an F_1 -score of only 0.959 due to low precision, whereas SCFDM_{part} improves it to 0.977. CFDMiner discovers a large number of spurious CFDs that hold only by coincidence rather than reflecting true attribute correlations. In comparison, SCFDM_{part} identifies correlated attributes and achieves high precision and F_1 -score in CFD discovery.

The effectiveness of SCFDM_{part} in discovering constant CFDs stems from its ability to extract correlated attribute sets, apply representative tuple sampling, and perform parallel CFD discovery on sub-tables, which reduces the search space and improves the rule quality of constant CFD discovery.

H.2 Parameter Sensitivity Results on the Remaining Five Datasets in Exp-4.

To supplement the parameter sensitivity evaluation presented in Exp-4, we report the complete experimental results regarding the impact of normalized support thresholds $|\hat{\sigma}|$ on the remaining five datasets (Census, Crop, Flight, Adult, and Syn-2) in Table 9.

The experimental results across these five datasets demonstrate a performance trend highly consistent with the findings observed on the RT and Syn-1 datasets in Exp-4. For the relatively simple datasets, (e.g., Flight and Adult), all baselines (CTANE, Itemset-First, and FD-First) complete their execution within the 9-hour time limit across support thresholds from 2×10^{-3} to 5×10^{-3} , thus matching the optimal F_1 -score achieved by SCFDM.

In contrast, on datasets with high tuple diversity or large scale, including Census, Crop, and Syn-2, the performance of all baselines degrades significantly when the support threshold is configured to a low value (e.g., $|\hat{\sigma}| = 10^{-3}$). Specifically, under $|\hat{\sigma}| = 10^{-3}$, the F_1 -scores of CTANE drop to 0.663 on Census, 0.588 on Crop, and 0.472 on Syn-2 due to time-limit termination. Conversely, SCFDM bypasses these computational bottlenecks by extracting correlated attribute sets and applying parallel CFD discovery on row-sampled sub-tables, maintaining high F_1 -scores (above 0.89) across all settings and outperforming baselines in low-support scenarios.

H.3 Impact of Maximum CFD Size and Heuristic Optimization

The worst-case runtime of CFD discovery is affected by the maximum CFD size $|k|$, as the search space grows exponentially with $\binom{m}{|k|}$. Instead of using a fixed limit on $|k|$, SCFDM adopts a heuristic strategy to determine $|k|$.

Specifically, our framework determines $|k|$ from the weight distribution in the global attention map $\tilde{\mathbf{A}}$. Instead of a fixed constraint on the maximum rule length $|k|$, the retention factor γ flexibly controls the maximum rule length k_Y for each attribute Y . For each consequent Y , SCFDM ranks candidate antecedents in descending order of attention weights $\tilde{\mathbf{A}}[Y, \cdot]$. The adaptive CFD size limit k_Y is then defined as the smallest number of top-ranked attributes whose cumulative attention mass reaches $\gamma \in (0, 1]$, i.e., $\sum_{i=1}^{k_Y} \tilde{\mathbf{A}}[Y, i] \geq \gamma$.

To ensure a theoretically grounded and automatic initialization

of the hyperparameter γ , we derive a robust formulation for selecting its optimal value, inspired by cumulative variance criteria in dimensionality reduction [59, 87]:

$$\gamma = \max \left(\gamma_0, 1 - \frac{1}{m} \sum_{i=1}^m \mathcal{H}(\bar{A}[A_i, \cdot]) \right),$$

where $\gamma_0 = 0.85$ represents the baseline information density, and m is the total number of attributes. Specifically, the normalized Shannon entropy [80] for each attention row vector is defined as:

$$\mathcal{H}(\bar{A}[Y, \cdot]) = -\frac{1}{\ln m} \sum_{k=1}^m \bar{A}[Y, k] \ln (\bar{A}[Y, k] + \epsilon_0),$$

where $\epsilon_0 = 10^{-5}$ is a tiny smoothing constant to prevent numerical underflow for zero entries. A low-entropy (highly concentrated) attention distribution indicates that a small number of antecedents carry most of the dependency signal, which increases γ and results in stricter selection that retains only strongly supported antecedents. Conversely, a high-entropy (near-uniform) distribution indicates that attention is spread more evenly across candidates, which decreases γ and restricts the antecedent set size, reducing combinatorial explosion under uncertain dependency patterns.

Our empirical evaluations across real-world relations (e.g., the Adult and Crop datasets) validate the robustness and effectiveness of this heuristic strategy. In our experiments, adjusting the retention factor γ via our formulation within the range of $[0.85, 0.95]$ consistently enables the framework to dynamically bound k_Y between 2 and 7 across diverse attribute schemas. The empirical results on the Crop dataset under the default setting show that this heuristic strategy, which adaptively determines a per-attribute limit k_Y instead of using a fixed limit k (e.g., $k = 6$) for all attributes, achieves a better balance between efficiency and discovery accuracy. It reduces the runtime from 1,724.2s to 655.4s and improves the F_1 -score from 98.0% to 98.2%. This data-driven filtering prunes unpromising attribute combinations, proving that our heuristic setting of k_Y reduces search space while maintaining discovery accuracy.

H.4 Systematic Guidance for Hyperparameter Optimization

To provide a workflow for deploying SCFDM, this section presents a hyperparameter optimization guide. The configuration of SCFDM is divided into two categories: (1) *discovery guarantee parameters*, which control sampling bounds; and (2) *structural partitioning parameters*, which determine correlated attribute sets.

Category 1: Discovery guarantee parameters ($\epsilon, \tau, \beta, \hat{\sigma}, \delta$). These parameters directly determine the sample size N in Theorem 2. Specifically, the normalized support $\hat{\sigma}$ and confidence δ are directly specified by the user to define the rule discovery criteria and mining granularity, with typical empirical values set to $\hat{\sigma} = 2 \times 10^{-3}$ and $\delta = 0.95$. For the remaining parameters, we recommend a sequential configuration starting with the recall parameter β . As the core control for rule preservation, β defines the sampling bounds and is typically set to $\beta \geq 0.90$ in data cleaning tasks to reduce the risk of missing CFDs. Subsequently, the additive support error ϵ is tuned, which increases the sample size N at rate $O(1/\epsilon^2)$. Users should initialize $\epsilon = 5 \times 10^{-4}$; for datasets with attribute count $m > 50$, it can be reduced to 2×10^{-4} to increase the sample

size and better preserve rare tuple patterns. Finally, the Chernoff ratio τ controls the selection of N in Theorem 2. Setting $\tau = 0.05$ by default provides tight sample bounds when the normalized support threshold $\hat{\sigma} \leq 0.14$. For relations with $\hat{\sigma} \leq 0.02$, τ can be reduced to 0.02 to enforce a tighter sampling.

Category 2: Structural partitioning parameters (γ, θ_{att}). These parameters dictate the pruning threshold within the correlation extraction stage (Section 6.3) and directly manipulate the sub-table candidate space. Their adjustment starts with the automation of the attention retention factor γ . Instead of setting a fixed threshold, users should compute γ following the formulation in Appendix H.3 with the base density parameter $\gamma_0 = 0.85$, which dynamically balances the local size k_Y (typically between 2 and 7). Subsequently, the scaling factor z in the significance threshold $\theta_{att} = z/m$ is adaptively formulated as: $z = \frac{1}{1 - \bar{\mathcal{H}} + \epsilon_1}$, where $\bar{\mathcal{H}} = \frac{1}{m} \sum_{i=1}^m \mathcal{H}(\bar{A}[A_i, \cdot])$ represents the global mean normalized Shannon entropy [80], and $\epsilon_1 = 10^{-5}$ is a numerical stability constant. On clean datasets with clear dependency signals, the system has low entropy ($\bar{\mathcal{H}} \rightarrow 0$), and z is reduced toward 1. As a result, the pruning threshold θ_{att} converges to the baseline $1/m$, preserving attributes that carry useful signals. In contrast, when noise is high and attention is widely spread ($\bar{\mathcal{H}} \rightarrow 1$), z increases, which raises θ_{att} and filters out weak candidate combinations. This prevents the search space from growing excessively without requiring fixed thresholds (θ_{att}).

Step-by-step optimization pipeline. For a new dataset, the operational sequence is executed as follows. First, the normalized support $\hat{\sigma}$ and confidence δ are user-specified parameters that define the rule discovery problem. The formal problem statement is given in Section 3 (p. 4). Based on our empirical settings, we initialize these sampling bounds with the default values $\hat{\sigma} = 2 \times 10^{-3}$, $\beta = 0.9$, $\delta = 0.95$. We set $\tau = 0.05$ by default, and reduce it to 0.02 if the normalized support threshold $\hat{\sigma} \leq 0.02$. We then set $\epsilon = 5 \times 10^{-4}$, and reduce it to 2×10^{-4} if the attribute count $m > 50$. Second, run preview profiling on a 5%–20% data slice to extract the global attention map $\bar{A}$ and compute the mean entropy $\bar{\mathcal{H}}$. Third, resolve γ and z directly from $\bar{\mathcal{H}}$. For example, on a dataset with $\bar{\mathcal{H}} = 0.20$, the formulation yields $z = 1.25$, preserving more candidate attributes. Conversely, on a noisy dataset with $\bar{\mathcal{H}} = 0.80$, it yields $z = 5.00$, leading to stronger pruning of the sub-table space.

H.5 Quantitative Evaluation of Data Repair in Exp-6

To support the repair claim in Exp-6, we evaluate repair performance on the Adult dataset, where the injected errors are known by construction. We introduce errors into 1,000 cells, including incorrect Gender values, missing values (marked as “?”), and corrupted Marital entries. Applying the discovered CFDs, including φ_5 and φ_6 , as repair rules corrects 912 of the 1,000 injected errors. This yields 94.6% repair precision, 91.2% recall, and a repair F_1 of 0.929. For φ_6 , which exactly matches Hours = 50, 31 repairs are triggered, of which 28 are correct, demonstrating its high repair accuracy. These results demonstrate that the discovered CFDs support effective data repair with high accuracy.

H.6 Additional Scalability Results on the Remaining Datasets

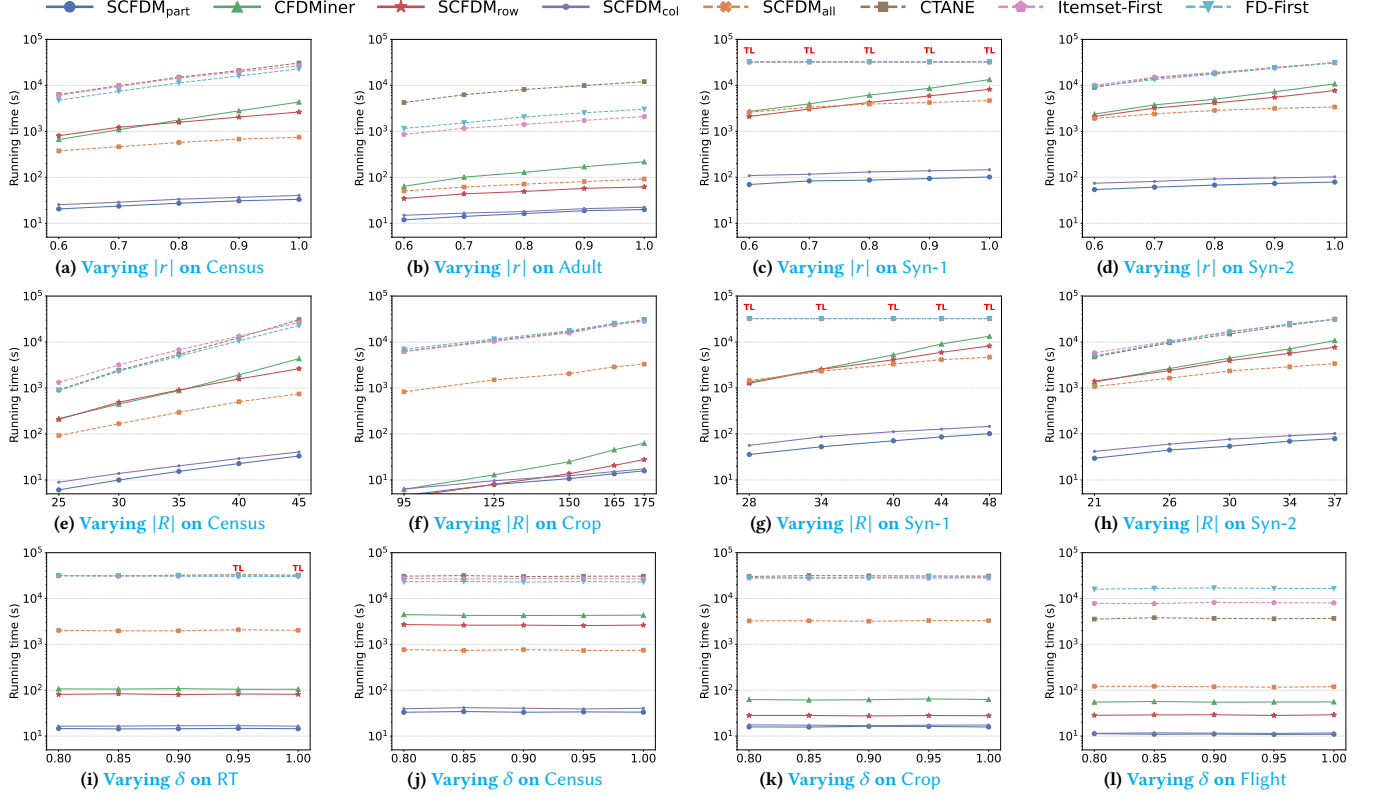


Figure 5: Additional scalability evaluation on remaining datasets (varying $|r|$, $|R|$, and δ)

Table 8: Parallel performance of SCFDM with varying q on the remaining three datasets

Algorithm	Dataset	Execution Time (s)			
		$q = 28$	$q = 56$	$q = 84$	$q = 112$
SCFDM _{part}	Crop	44.75	32.19	22.85	15.80
SCFDM _{all}	Crop	10,156.74	6,008.17	4,180.84	3,293.83
SCFDM _{part}	Flight	34.76	26.67	17.02	10.98
SCFDM _{all}	Flight	309.77	217.01	142.21	119.32
SCFDM _{part}	Syn-2	252.26	168.80	113.75	78.96
SCFDM _{all}	Syn-2	9,306.72	6,399.92	4,468.14	3,392.30

Following the discussion in Section 9.2 (Exp-2), we report the scalability results for the datasets omitted from Figure 4 and Table 3 in the main text. All experiments follow the default settings of Section 9.1 and use the same eight algorithms. The trends are consistent with those presented in the main text: SCFDM scales gradually with each parameter, whereas the baselines grow exponentially and frequently exceed the time limit on wider or denser schemas.

Varying $|r|$. Figures 5(a)–5(d) show the impact of tuning the tuple sampling ratio $|r|$ from 0.6 to 1.0 on the remaining four datasets (Census, Adult, Syn-1, and Syn-2). As $|r|$ increases, the running times of all algorithms rise due to the expanded tuple validation workload. However, while baseline miners (e.g., CTANE, Itemset-First, and FD-First) scale sharply and frequently hit the 9-hour timeout limit (32,400 s) on the synthetic dataset Syn-1, both SCFDM_{part} and SCFDM_{all} exhibit near-linear, graceful runtime growth across all settings, matching the scalability trends observed

on Crop, Flight, and RT in Figure 4.

Varying $|R|$. Figures 5(e)–5(h) illustrate the running time when scaling the attribute count $|R|$ across medium schemas (Census, 25–45; Syn-2, 21–37), an ultra-wide schema (Crop, 95–175), and a synthetic schema (Syn-1, 28–48). Due to the combinatorial explosion of the candidate search lattice, the execution times of baseline miners escalate exponentially and reach the time limit on wide or dense relations (e.g., Syn-1). In contrast, AttrFinder prunes uncorrelated attributes before mining, allowing SCFDM_{part} and SCFDM_{all} to maintain gradual runtime increases throughout all schema widths.

Varying δ . Figures 5(i)–5(l) show the sensitivity under confidence threshold $\delta \in [0.80, 1.00]$ on the four omitted real-world datasets (RT, Census, Crop, and Flight). Because δ only prunes candidate CFDs during validation without materially reducing the candidate generation space, the execution times of all tested algorithms remain nearly flat across the entire range of δ , consistently echoing the patterns observed on Syn-2 and Adult in Figure 4.

Varying q . Table 8 extends the parallel scalability evaluation (Table 3) to Crop, Flight, and Syn-2 by scaling the number of CPU cores q from 28 to 112. Across all three datasets, SCFDM achieves steady and near-linear performance acceleration as computational resources scale. Specifically, when quadrupling the core count from 28 to 112, SCFDM_{part} achieves speedups of 2.83 $\times$ (Crop), 3.17 $\times$ (Flight), and 3.19 $\times$ (Syn-2), while SCFDM_{all} delivers speedups of 3.08 $\times$, 2.60 $\times$, and 2.74 $\times$, respectively. These results demonstrate that decomposing relations into independent sub-tables effectively improves the efficiency of data-parallel CFD discovery.

Table 9: Parameter sensitivity results (support threshold $|\hat{\sigma}|$) on the remaining five datasets in Exp-4

Dataset	Algorithm	Metric	1×10^{-3}	2×10^{-3}	3×10^{-3}	4×10^{-3}	5×10^{-3}
Census	CFDMiner		0.103	0.117	0.145	0.160	0.188
	CTANE		0.663	0.971	0.984	0.979	0.988
	Itemset-First F_1		0.755	0.973	0.981	0.986	0.983
	FD-First		0.712	0.978	0.981	0.983	0.983
	SCFDM		0.919	0.985	0.982	0.986	0.988
Crop	CFDMiner		0.094	0.155	0.157	0.168	0.172
	CTANE		0.588	0.950	0.958	0.964	0.975
	Itemset-First F_1		0.674	0.949	0.959	0.968	0.972
	FD-First		0.695	0.949	0.970	0.970	0.970
	SCFDM		0.898	0.952	0.974	0.970	0.972
Flight	CFDMiner		0.140	0.246	0.233	0.251	0.256
	CTANE		0.863	0.997	0.996	0.993	0.996
	Itemset-First F_1		0.907	0.997	0.997	0.995	0.996
	FD-First		0.924	0.997	0.996	0.998	0.998
	SCFDM		0.994	0.997	0.998	0.995	0.998
Adult	CFDMiner		0.165	0.505	0.512	0.537	0.565
	CTANE		0.988	1.000	1.000	0.996	1.000
	Itemset-First F_1		0.992	1.000	0.998	0.997	0.998
	FD-First		0.994	1.000	0.994	0.995	1.000
	SCFDM		0.994	1.000	0.997	1.000	1.000
Syn-2	CFDMiner		0.087	0.146	0.166	0.174	0.178
	CTANE		0.472	0.970	0.970	0.975	0.990
	Itemset-First F_1		0.504	0.967	0.968	0.977	0.983
	FD-First		0.521	0.965	0.970	0.980	0.983
	SCFDM		0.931	0.973	0.972	0.985	0.990

* Results obtained before reaching the resource limit (9 hours of time or 224GB of memory).
Bold indicates the best performance.

H.7 Computation reduction of AttrFinder

The computation reduction in AttrFinder is the ratio of the lattice traversal computational overhead in the sub-tables to that of the original instance r , excluding representative sampling and pruning. It can be expressed as: $(\sum_{i=1}^l m_i \times d^{m_i}) / (m \times d^m)$, where m is the number of attributes in r , d is the average domain size, l is the number of sub-tables, and m_i is the number of attributes in the i -th sub-table. In the experiment, AttrFinder identified 85, 45, 175, 25, 14, 48, and 37 correlated vector sets on the RT, Census, Crop, Flight, Adult, Syn-1, and Syn-2 datasets, respectively. Compared to direct mining, this leads to significant computational overhead reductions of at least $\frac{40}{d_{RT}^{45}}$, $\frac{12}{d_{Census}^{23}}$, $\frac{70}{d_{Crop}^{105}}$, $\frac{11}{d_{Flight}^{44}}$, $\frac{8}{d_{Adult}^8}$, $\frac{20}{d_{Syn-1}^{28}}$, and $\frac{19}{d_{Syn-2}^{18}}$ on these datasets (calculated based on the largest sub-table).

H.8 Computation reduction of RepSampler

The computation reduction in RepSampler is the ratio of the number of tuples in the sample (n_s) to the total number of tuples in the original relation (n), i.e., n_s/n . RepSampler selected 8,519, 10,544, 16,063, 41,057, 8,422, 8,408, and 8,397 representative tuples from the RT, Census, Crop, Flight, Adult, Syn-1, and Syn-2 datasets, respectively. This selection yielded computation reductions of 0.069, 0.053, 0.049, 0.041, 0.177, 0.420, and 0.420, respectively. Notably, the computational overhead exhibits a linear decrease as the sample size n_s is reduced.

H.9 Details of Experimental Real-world Datasets

We provide a comprehensive overview of the five real world datasets used in our experiments, focusing on their structural characteristics and relevance to CFD discovery.

(1) RT-IOT (RT). The RT-IOT-2022 dataset is a comprehensive resource derived from a real time IOT infrastructure [81]. Unlike traditional static datasets, it integrates a diverse range of physical IOT devices including ThingSpeak LED, Wipro Bulb, and MQTT Temp alongside sophisticated network attack methodologies. The dataset is designed to provide a general representation of real world scenarios by capturing both normal operations and adversarial behaviors. To ensure high fidelity in network traffic analysis, the bidirectional attributes of the traffic were meticulously captured using the Zeek network monitoring tool and the Flowmeter plugin. This process resulted in a rich feature set that allows researchers to analyze complex traffic patterns across various protocols.

The dataset contains a total of 123,117 instances, categorized into attack patterns and normal patterns as shown in Table 10. In our research, this dataset serves as a benchmark for mining CFDs. Specifically, the coexistence of normal and malicious traffic allows us to identify dependencies that hold under normal conditions but are violated during specific attack phases. For example, a CFD may capture the stable relationship between a device type and its expected traffic volume, which becomes an essential constraint for identifying anomalous behavior in IOT environments.

(2) Census-Income (Census). The Census-Income dataset consists of weighted census records extracted from the 1994 and 1995 Current Population Surveys conducted by the U.S. Census Bureau. This dataset is a high dimensional resource containing 41 demographic and employment related variables for each individual, including features such as age, class of worker, education, and marital status. It provides a total of 199,523 training instances and 99,762 test instances, which were split in approximately a two thirds to one third proportion. A critical attribute of this dataset is the instance weight, which indicates the number of people in the actual population that each record represents due to stratified sampling. While this field is essential for deriving statistical conclusions about the general population, it is typically excluded from the feature set used in classification models. The primary task associated with this dataset is a binary classification problem. Similar to the original Adult database, incomes have been binned at the 50,000 dollar level. However, the target variable in this specific version was derived from the total person income field rather than adjusted gross income, which introduces distinct statistical characteristics.

For the purpose of mining CFDs, the Census Income dataset is particularly valuable due to its rich set of categorical and numerical attributes. It allows for the discovery of complex dependencies between demographic profiles and socio economic status, such as the relationship between occupation codes and wage per hour conditioned on specific industry sectors.

(3) Crop mapping using fused optical radar data (Crop). The Crop dataset is a fused bi temporal resource designed for sophisticated cropland classification near Winnipeg, Manitoba, Canada. Collected in 2012, this dataset combines optical imagery from Rapid-Eye satellites with Polarimetric Synthetic Aperture Radar (PolSAR) data from the Unmanned Aerial Vehicle Synthetic Aperture Radar (UAVSAR) system. By integrating these complementary sensing modalities, the dataset captures a significant number of temporal, spectral, textural, and polarimetric features that are essential for distinguishing between various crop types.

The dataset contains 175 attributes in total, structured around

Table 10: Distribution of attack and normal patterns in the RT-IOT dataset.

Attack Patterns	Count	Normal Patterns	Count
DOS SYN Hping	94,659	Amazon Alexa	86,842
ARP poisoning	7,750	MQTT	8,108
NMAP UDP SCAN	2,590	Thing speak	4,146
NMAP XMAS TREE SCAN	2,010	Wipro bulb	253
NMAP OS DETECTION	2,000		
NMAP TCP scan	1,002		
DDOS Slowloris	534		
Metasploit Brute Force SSH	37		
NMAP FIN SCAN	28		

two specific observation dates: July 05 and July 14, 2012. The feature space is divided into several categories: (1) Polarimetric features: 49 radar features per date (f1 to f98) capturing scattering properties such as sigHH, sigVV, and complex decompositions like Cloude Pottier parameters (Entropy, Anisotropy) and Freeman Durden components. (2) Optical features: 38 optical features per date (f99 to f174) including spectral bands (Blue, Green, Red, Red edge, NIR) and derived vegetation indices such as NDVI, EVI, and SAVI, alongside textural metrics derived from Principal Component Analysis. There are seven distinct crop type classes identified in this dataset: Corn, Peas, Canola, Soybeans, Oats, Wheat, and Broadleaf.

In our study, the Crop dataset serves as a benchmark for mining CFDs across multi modal sensor data. The rich set of 175 attributes allows for the discovery of dependencies where spectral signatures from optical sensors are constrained by polarimetric backscatter measurements. For instance, a CFD might define a stable relationship between a specific crop type and its expected NDVI range, conditioned on the stability of its radar backscatter cross section during different growth stages.

(4) 2015 Flight Delays and Cancellations (Flight). The Flight dataset, specifically the 2015 Flight Delays and Cancellations resource, provides an exhaustive record of domestic flight operations within the United States. It encompasses nearly 5.8 million flights, capturing critical metrics such as scheduled and actual departure times, arrival delays, and cancellation reasons. The dataset is structured to facilitate multi dimensional analysis across three primary entities: Airlines (14 major carriers), Airports (322 locations), and Time (year, month, day, and day of week). The raw data underwent a rigorous cleaning process to ensure analytical integrity. This included converting blank entries into

NA values, omitting records with missing scheduled time or air time, and merging flight records with airport and airline lookup tables to resolve numeric codes into descriptive names. Statistical profiling reveals that approximately 1.6 percent of all flights were cancelled, with over half of these cancellations attributed to weather conditions (category B), followed by airline related issues (category A) and air system delays (category C).

For CFD discovery, this dataset is particularly valuable due to the intricate relationships between categorical attributes (e.g., Origin Airport, Airline) and numerical performance metrics (e.g., Departure Delay, Taxi In time). It allows for the identification of dependencies where the probability of a delay is constrained by specific temporal or geographical contexts. For example, a CFD might capture a stable constraint where flights operated by a specific carrier from a certain hub consistently maintain a lower taxi out duration, provided the weather conditions remain within normal parameters.

(5) Adult. The Adult dataset was extracted by Barry Becker from the 1994 United States Census database. To ensure data quality, a set of reasonably clean records was filtered based on specific demographic and economic conditions, including individuals older than 16 years, an adjusted gross income exceeding 100 dollars, and a recorded work volume of more than zero hours per week.

The primary prediction task for this dataset is a binary classification problem: determining whether an individual’s annual income exceeds the 50,000 dollar threshold. The dataset comprises 14 diverse attributes that capture a holistic view of an individual’s socio economic profile: (1) Categorical Attributes: These include workclass (e.g., Private, Federal gov), education level, marital status, occupation, relationship role, race, sex, and native country. (2) Continuous Attributes: These encompass age, final weight (fnlwgt), education num, capital gain, capital loss, and hours per week.

In the context of our research on CFDs, the Adult dataset provides a robust environment for discovering complex logical constraints. Its mix of categorical and continuous variables allows for the identification of dependencies where professional attributes (such as occupation and education) are functionally linked to income levels under specific demographic conditions (such as work-class or age groups). For instance, a CFD might define a high confidence dependency between a doctorate degree and the high income class, specifically within the context of private sector employment.